

Premier University, Chittagong



FINAL PROJECT REPORT

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1. Executive Summary

This project demonstrates how Hadoop-based distributed processing can be applied to a large historical Mutual Funds dataset. The dataset contains fund-house information, scheme classifications, scheme names, observation dates, and Net Asset Values (NAV). The work uses HDFS for distributed storage, Hadoop MapReduce for Java-based batch analytics, and Apache Pig for higher-level data-flow analysis.

The project contains three MapReduce experiments and five Pig experiments. Together, they cover maximum-value analysis, averages, categorical counts, filtering, and grouped statistics. The execution was performed in a Windows-based Hadoop environment using HDFS and YARN, with Apache Pig running in MapReduce mode.

A total of 29,033,647 records were read from the Mutual Funds CSV during Pig processing. The input file was stored in HDFS at /mutual_funds/Mutual_Funds.csv. The experiments were completed successfully and produced outputs such as a maximum NAV of 1,918,470.3931, 71 Fund House groups in the Pig maximum-NAV analysis, 84 Scheme Category groups in the average-NAV analysis, and 78 groups in the Scheme-Type record-count analysis.

2. Project Objectives

Set up Hadoop services and store the Mutual Funds dataset in HDFS.

Use MapReduce to compute the maximum NAV, average NAV by Fund House, and Scheme Category record counts.

Use Apache Pig to load, filter, group, and aggregate the same dataset.

Compare the usefulness of low-level MapReduce processing with higher-level Pig Latin data flows.

Document the execution process, results, observations, and project repository for reproducibility.

3. Dataset Description

The project uses a CSV dataset named “<https://www.kaggle.com/datasets/balajisr/indian-mutual-funds-dataset-2023>” containing historical mutual-fund observations. Each row represents a scheme observation with a fund house, scheme type, category, code, scheme name, date, and NAV.

Attribute	Type	Description
Fund_House	Text	Name of the mutual-fund company or fund house
Scheme_Type	Text	Type of mutual-fund scheme
Scheme_Category	Text	Category or classification of the scheme
Scheme_Code	Text	Scheme identifier/code
Scheme_Name	Text	Complete name of the mutual-fund scheme
Date	Text/Date	Date of the NAV observation
NAV	Numeric	Net Asset Value used for filtering and aggregation

Local dataset path:

C:\Mutual_Funds.csv

HDFS dataset path:

/mutual_funds/Mutual_Funds.csv

4. Technology Stack and Environment

Technology	Role in the project
Apache Hadoop	Distributed processing framework
HDFS	Storage of the large CSV dataset
YARN	Cluster resource management for MapReduce jobs
Hadoop MapReduce	Java-based distributed batch analysis
Apache Pig 0.18.0	Pig Latin data-flow analysis
Java	Implementation language for MapReduce programs
Windows Command Prompt	Execution environment used in the laboratory

The observed execution environment used Hadoop 3.2.4 and Apache Pig 0.18.0. The Hadoop services were started using the Windows command scripts and checked with jps before data processing.

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26280.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>cd C:\hadoop\sbin
C:\hadoop\sbin>start-dfs.cmd
C:\hadoop\sbin>start-yarn.cmd
starting yarn daemons
C:\hadoop\sbin>jps
11968 Jps
14256 NameNode
7984 NodeManager
2696 DataNode
6988 ResourceManager
C:\hadoop\sbin>C:\Mutual_Funds.csv
C:\hadoop\sbin>hdfs dfs -mkdir /mutual_funds
C:\hadoop\sbin>hdfs dfs -put "C:\Mutual_Funds.csv" /mutual_funds/
C:\hadoop\sbin>hdfs dfs -ls /mutual_funds
Found 1 items
-rw-r--r-- 1 ASUS supergroup 4459429272 2026-09-11 18:27 /mutual_funds/Mutual_Funds.csv
C:\hadoop\sbin>hdfs dfs -cat /mutual_funds/Mutual_Funds.csv | head
'head' is not recognized as an internal or external command,
operable program or batch file.
C:\hadoop\sbin>hdfs dfs -ls -h /mutual_funds
Found 1 items
-rw-r--r-- 1 ASUS supergroup 4.2 G 2026-09-11 18:27 /mutual_funds/Mutual_Funds.csv
C:\hadoop\sbin>
```

5. Processing Architecture

The overall workflow begins with a local CSV file. The file is uploaded to HDFS, where Hadoop provides distributed storage. MapReduce jobs read the HDFS input and generate aggregated outputs in HDFS output directories. Pig jobs read the same HDFS dataset and express filtering/grouping/aggregation operations using Pig Latin.

6. MapReduce Analysis

Three MapReduce experiments were used to demonstrate distributed key-value processing over the Mutual Funds dataset. Each experiment follows the standard Driver → Mapper → Shuffle/Sort → Reducer → HDFS output pattern.

6.1 Maximum NAV

Objective: determine the highest NAV value present in the dataset. The Mapper extracts valid NAV values, while the Reducer compares values and retains the maximum.

```
import java.io.IOException;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.DoubleWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class MaxNAV {

    public static class MaxMapper
    extends Mapper<LongWritable, Text, Text, DoubleWritable> {

        private final Text outKey = new Text("Maximum NAV");

        @Override
        public void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException {
```

```

String line = value.toString().trim();

if (line.isEmpty() || line.startsWith("Fund_House,")) {
    return;
}

String[] fields = line.split(",", -1);

if (fields.length < 7) {
    return;
}

try {
    double nav = Double.parseDouble(fields[6].trim());
    context.write(outKey, new DoubleWritable(nav));
} catch (NumberFormatException e) {
    // Ignore invalid NAV values.
}
}
}

public static class MaxReducer
    extends Reducer<Text, DoubleWritable, Text, DoubleWritable> {

    @Override
    public void reduce(Text key, Iterable<DoubleWritable> values,
        Context context)
        throws IOException, InterruptedException {

        double max = Double.NEGATIVE_INFINITY;

        for (DoubleWritable value : values) {
            max = Math.max(max, value.get());
        }

        if (max != Double.NEGATIVE_INFINITY) {
            context.write(key, new DoubleWritable(max));
        }
    }
}

```

```
}  
}
```

```
public static void main(String[] args) throws Exception {
```

```
    if (args.length != 2) {  
        System.err.println(  
            "Usage: MaxNAV <input path> <output path>");  
        System.exit(2);  
    }
```

```
    Configuration conf = new Configuration();  
    Job job = Job.getInstance(conf, "Maximum NAV");
```

```
    job.setJarByClass(MaxNAV.class);  
    job.setMapperClass(MaxMapper.class);  
    job.setReducerClass(MaxReducer.class);
```

```
    job.setMapOutputKeyClass(Text.class);  
    job.setMapOutputValueClass(DoubleWritable.class);  
    job.setOutputKeyClass(Text.class);  
    job.setOutputValueClass(DoubleWritable.class);
```

```
    job.setNumReduceTasks(1);
```

```
    FileInputFormat.addInputPath(job, new Path(args[0]));  
    FileOutputFormat.setOutputPath(job, new Path(args[1]));
```

```
    System.exit(job.waitForCompletion(true) ? 0 : 1);  
}  
}
```

The completed job reported a maximum NAV of 1,918,470.3931.

```
Administrator: Command Prompt
C:\hadoop\sbin>mkdir C:\mutual_funds
C:\hadoop\sbin>cd C:\mutual_funds\java
C:\mutual_funds\java>notepad MaxNAV.java
C:\mutual_funds\java>javac -classpath "%HADOOP_HOME%\share\hadoop\common\*;%HADOOP_HOME%\share\hadoop\common\lib\*;%HADOOP_HOME%\share\hadoop\mapreduce\*;%HADOOP_HOME%\share\hadoop\mapreduce\lib\*" MaxNAV.java
C:\mutual_funds\java>jar -cxf MaxNAV.jar *.class
added manifest
adding: MaxNAV$Mapper.class(in = 1902) (out= 788)(deflated 58%)
adding: MaxNAV$MaxReducer.class(in = 1749) (out= 747)(deflated 57%)
adding: MaxNAV.class(in = 1434) (out= 793)(deflated 44%)
C:\mutual_funds\java>hdfs dfs -rm -r /output/maxnav
rm: '/output/maxnav': No such file or directory
C:\mutual_funds\java>hadoop jar MaxNAV.jar MaxNAV /mutual_funds/Mutual_Funds.csv /output/maxnav
2026-09-11 18:31:32,600 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 18:31:33,096 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-11 18:31:33,114 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/ASUS/staging/job_1789129568662_0001
2026-09-11 18:31:33,349 INFO input.FileInputFormat: Total input files to process : 1
2026-09-11 18:31:33,443 INFO mapreduce.JobSubmitter: number of splits:34
2026-09-11 18:31:33,572 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789129568662_0001
2026-09-11 18:31:33,574 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-11 18:31:33,746 INFO conf.Configuration: resource-types.xml not found
2026-09-11 18:31:33,747 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-11 18:31:34,250 INFO impl.YarnClientImpl: Submitted application application_1789129568662_0001
2026-09-11 18:31:34,381 INFO mapreduce.Job: the url to track the job: http://HOU:8088/proxy/application_1789129568662_0001/
2026-09-11 18:31:34,382 INFO mapreduce.Job: Running Job: job_1789129568662_0001
2026-09-11 18:31:42,499 INFO mapreduce.Job: Job job_1789129568662_0001 running in uber mode : false
2026-09-11 18:31:42,500 INFO mapreduce.Job: map 0% reduce 0%
2026-09-11 18:31:58,922 INFO mapreduce.Job: map 12% reduce 0%
2026-09-11 18:31:59,928 INFO mapreduce.Job: map 18% reduce 0%
2026-09-11 18:32:12,210 INFO mapreduce.Job: map 20% reduce 0%
2026-09-11 18:32:13,234 INFO mapreduce.Job: map 32% reduce 0%
2026-09-11 18:32:18,300 INFO mapreduce.Job: map 35% reduce 0%
2026-09-11 18:32:23,379 INFO mapreduce.Job: map 41% reduce 0%
2026-09-11 18:32:24,394 INFO mapreduce.Job: map 44% reduce 0%
2026-09-11 18:32:27,431 INFO mapreduce.Job: map 47% reduce 0%
2026-09-11 18:32:30,486 INFO mapreduce.Job: map 50% reduce 10%
2026-09-11 18:32:32,512 INFO mapreduce.Job: map 59% reduce 16%
2026-09-11 18:32:35,567 INFO mapreduce.Job: map 62% reduce 16%
2026-09-11 18:32:36,584 INFO mapreduce.Job: map 62% reduce 21%
2026-09-11 18:32:39,632 INFO mapreduce.Job: map 65% reduce 21%
2026-09-11 18:32:40,680 INFO mapreduce.Job: map 68% reduce 21%
2026-09-11 18:32:41,696 INFO mapreduce.Job: map 74% reduce 21%
```

```
Administrator: Command Prompt
2026-09-11 18:32:47,851 INFO mapreduce.Job: map 76% reduce 25%
2026-09-11 18:32:48,906 INFO mapreduce.Job: map 79% reduce 25%
2026-09-11 18:32:54,084 INFO mapreduce.Job: map 85% reduce 25%
2026-09-11 18:32:55,139 INFO mapreduce.Job: map 85% reduce 26%
2026-09-11 18:32:58,189 INFO mapreduce.Job: map 88% reduce 26%
2026-09-11 18:33:01,241 INFO mapreduce.Job: map 91% reduce 29%
2026-09-11 18:33:03,276 INFO mapreduce.Job: map 97% reduce 29%
2026-09-11 18:33:04,300 INFO mapreduce.Job: map 100% reduce 29%
2026-09-11 18:33:07,349 INFO mapreduce.Job: map 100% reduce 32%
2026-09-11 18:33:13,423 INFO mapreduce.Job: map 100% reduce 75%
2026-09-11 18:33:19,500 INFO mapreduce.Job: map 100% reduce 92%
2026-09-11 18:33:22,535 INFO mapreduce.Job: map 100% reduce 100%
2026-09-11 18:33:22,546 INFO mapreduce.Job: Job job_1789129568662_0001 completed successfully
2026-09-11 18:33:22,693 INFO mapreduce.Job: Counters: 55
File System Counters
  FILE: Number of bytes read=406378600
  FILE: Number of bytes written=421111021
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=4459568384
  HDFS: Number of bytes written=25
  HDFS: Number of read operations=107
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0
Job Counters
  Killed map tasks=1
  Launched map tasks=34
  Launched reduce tasks=1
  Data-local map tasks=34
  Total time spent by all maps in occupied slots (ms)=387787
  Total time spent by all reduces in occupied slots (ms)=68853
  Total time spent by all map tasks (ms)=387787
  Total time spent by all reduce tasks (ms)=68853
  Total vcore-milliseconds taken by all map tasks=387787
  Total vcore-milliseconds taken by all reduce tasks=68853
  Total megabyte-milliseconds taken by all map tasks=397093888
  Total megabyte-milliseconds taken by all reduce tasks=70505472
Map-Reduce Framework
  Map input records=29033647
  Map output records=29027042
  Map output bytes=348324504
  Map output materialized bytes=406378792
  Input split bytes=3944
  Combine input records=0
  Combine output records=0
  Reduce input groups=1
  Reduce shuffle bytes=406378792
```

```

CPU time spent (ms)=138710
Physical memory (bytes) snapshot=16079433728
Virtual memory (bytes) snapshot=24494600192
Total committed heap usage (bytes)=18427674624
Peak Map Physical memory (bytes)=503447552
Peak Map Virtual memory (bytes)=761020416
Peak Reduce Physical memory (bytes)=824180736
Peak Reduce Virtual memory (bytes)=1000816640

Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0

File Input Format Counters
  Bytes Read=4459564440
File Output Format Counters
  Bytes Written=25

C:\mutual_funds\java>hdfs dfs -cat /output/maxnav/part-r-00000
Maximum NAV      1918470.3931

C:\mutual_funds\java>

```

6.2 Average NAV by Fund House

Objective: calculate the average NAV associated with every Fund House. The Mapper emits Fund_House as the key and NAV as the value; the Reducer computes the mean for each group.

```

import java.io.IOException;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.DoubleWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class AverageNAV {

    public static class AvgMapper
    extends Mapper<LongWritable, Text, Text, DoubleWritable> {

        private final Text outKey = new Text();

```

```

@Override
public void map(LongWritable key, Text value, Context context)
throws IOException, InterruptedException {

    String line = value.toString().trim();

    if (line.isEmpty() || line.startsWith("Fund_House,")) {
        return;
    }

    String[] fields = line.split(",", -1);

    if (fields.length < 7) {
        return;
    }

    try {
        String fundHouse = fields[0].trim();
        double nav = Double.parseDouble(fields[6].trim());

        if (!fundHouse.isEmpty()) {
            outKey.set(fundHouse);
            context.write(outKey, new DoubleWritable(nav));
        }
    } catch (NumberFormatException e) {
        // Ignore invalid NAV values.
    }
}

public static class AvgReducer
extends Reducer<Text, DoubleWritable, Text, DoubleWritable> {

    @Override
    public void reduce(Text key, Iterable<DoubleWritable> values,
        Context context)
        throws IOException, InterruptedException {

```

```

double sum = 0.0;
long count = 0;

for (DoubleWritable value : values) {
    sum += value.get();
    count++;
}

if (count > 0) {
    double average = sum / count;
    context.write(key, new DoubleWritable(average));
}
}
}

public static void main(String[] args) throws Exception {

    if (args.length != 2) {
        System.err.println(
            "Usage: AverageNAV <input path> <output path>");
        System.exit(2);
    }

    Configuration conf = new Configuration();
    Job job = Job.getInstance(conf, "Average NAV by Fund House");

    job.setJarByClass(AverageNAV.class);
    job.setMapperClass(AvgMapper.class);
    job.setReducerClass(AvgReducer.class);

    job.setMapOutputKeyClass(Text.class);
    job.setMapOutputValueClass(DoubleWritable.class);
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(DoubleWritable.class);

    job.setNumReduceTasks(5);

    FileInputFormat.addInputPath(job, new Path(args[0]));

```

```
FileOutputFormat.setOutputPath(job, new Path(args[1]));
```

```
System.exit(job.waitForCompletion(true) ? 0 : 1);
```

```
}  
}
```

The MapReduce job produced a grouped output containing Fund House names and their average NAV values. The output file contained 2,778 bytes, and the successful execution used 34 map tasks.

```
Administrator: Command Prompt  
C:\mutual_funds\java>notepad AverageNAV.java  
C:\mutual_funds\java>javac -classpath "%HADOOP_HOME%\share\hadoop\common\*;%HADOOP_HOME%\share\hadoop\common\lib\*;%HADOOP_HOME%\share\hadoop\mapreduce\*;%HADOOP_HOME%\share\hadoop\mapreduce\lib\*" AverageNAV.java  
C:\mutual_funds\java>jar -cvf AverageNAV.jar *.class  
added manifest  
adding: AverageNAVAvgMapper.class(in = 1915) (out= 794)(deflated 58%)  
adding: AverageNAVAvgReducer.class(in = 1698) (out= 727)(deflated 57%)  
adding: AverageNAV.class(in = 1464) (out= 886)(deflated 44%)  
adding: MaxNAVAvgMapper.class(in = 1902) (out= 788)(deflated 58%)  
adding: MaxNAVAvgReducer.class(in = 1749) (out= 747)(deflated 57%)  
adding: MaxNAV.class(in = 1434) (out= 793)(deflated 44%)  
C:\mutual_funds\java>hdfs dfs -rm -r /output/avgnav  
rm: '/output/avgnav': No such file or directory  
C:\mutual_funds\java>hadoop jar AverageNAV.jar AverageNAV /mutual_funds/Mutual_Funds.csv /output/avgnav  
2026-09-11 18:37:34,254 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032  
2026-09-11 18:37:34,693 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.  
2026-09-11 18:37:34,707 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/ASUS/.staging/job_1789129568662_0002  
2026-09-11 18:37:34,857 INFO input.FileInputFormat: Total input files to process : 1  
2026-09-11 18:37:34,929 INFO mapreduce.JobSubmitter: number of splits:34  
2026-09-11 18:37:35,020 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789129568662_0002  
2026-09-11 18:37:35,021 INFO mapreduce.JobSubmitter: Executing with tokens: []  
2026-09-11 18:37:35,137 INFO conf.Configuration: resource-types.xml not found  
2026-09-11 18:37:35,137 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.  
2026-09-11 18:37:35,186 INFO impl.YarnClientImpl: Submitted application application_1789129568662_0002  
2026-09-11 18:37:35,208 INFO mapreduce.Job: The url to track the job: http://MDU:8088/proxy/application_1789129568662_0002/  
2026-09-11 18:37:35,208 INFO mapreduce.Job: Running job: job_1789129568662_0002  
2026-09-11 18:37:41,348 INFO mapreduce.Job: Job job_1789129568662_0002 running in uber mode : false  
2026-09-11 18:37:57,947 INFO mapreduce.Job: map 0% reduce 0%  
2026-09-11 18:37:59,154 INFO mapreduce.Job: map 15% reduce 0%  
2026-09-11 18:38:12,562 INFO mapreduce.Job: map 26% reduce 0%  
2026-09-11 18:38:13,631 INFO mapreduce.Job: map 32% reduce 0%  
2026-09-11 18:38:19,764 INFO mapreduce.Job: map 35% reduce 0%  
2026-09-11 18:38:25,914 INFO mapreduce.Job: map 38% reduce 0%  
2026-09-11 18:38:26,938 INFO mapreduce.Job: map 44% reduce 0%  
2026-09-11 18:38:30,996 INFO mapreduce.Job: map 47% reduce 0%  
2026-09-11 18:38:32,017 INFO mapreduce.Job: map 47% reduce 10%  
2026-09-11 18:38:35,000 INFO mapreduce.Job: map 50% reduce 16%  
2026-09-11 18:38:38,144 INFO mapreduce.Job: map 53% reduce 17%  
2026-09-11 18:38:39,163 INFO mapreduce.Job: map 59% reduce 17%  
2026-09-11 18:38:44,446 INFO mapreduce.Job: map 59% reduce 20%  
2026-09-11 18:38:48,537 INFO mapreduce.Job: map 62% reduce 20%  
2026-09-11 18:38:50,580 INFO mapreduce.Job: map 65% reduce 21%  
2026-09-11 18:38:53,620 INFO mapreduce.Job: map 74% reduce 21%
```

```

Administrator: Command Prompt
2026-09-11 18:38:56,716 INFO mapreduce.Job: map 74% reduce 25%
2026-09-11 18:39:00,839 INFO mapreduce.Job: map 76% reduce 25%
2026-09-11 18:39:02,887 INFO mapreduce.Job: map 79% reduce 25%
2026-09-11 18:39:06,963 INFO mapreduce.Job: map 80% reduce 25%
2026-09-11 18:39:09,009 INFO mapreduce.Job: map 88% reduce 29%
2026-09-11 18:39:14,077 INFO mapreduce.Job: map 94% reduce 30%
2026-09-11 18:39:15,091 INFO mapreduce.Job: map 97% reduce 30%
2026-09-11 18:39:16,102 INFO mapreduce.Job: map 100% reduce 30%
2026-09-11 18:39:20,142 INFO mapreduce.Job: map 100% reduce 71%
2026-09-11 18:39:26,220 INFO mapreduce.Job: map 100% reduce 84%
2026-09-11 18:39:32,290 INFO mapreduce.Job: map 100% reduce 96%
2026-09-11 18:39:35,327 INFO mapreduce.Job: map 100% reduce 100%
2026-09-11 18:39:35,333 INFO mapreduce.Job: Job job_1789129568662_0002 completed successfully
2026-09-11 18:39:35,461 INFO mapreduce.Job: Counters: 55
  File System Counters
    FILE: Number of bytes read=919961186
    FILE: Number of bytes written=1848276957
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=4459568384
    HDFS: Number of bytes written=2778
    HDFS: Number of read operations=107
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=2
    HDFS: Number of bytes read erasure-coded=0
  Job Counters
    Killed map tasks=1
    Launched map tasks=34
    Launched reduce tasks=1
    Data-local map tasks=34
    Total time spent by all maps in occupied slots (ms)=452758
    Total time spent by all reduces in occupied slots (ms)=81862
    Total time spent by all map tasks (ms)=452758
    Total time spent by all reduce tasks (ms)=81862
    Total vcore-milliseconds taken by all map tasks=452758
    Total vcore-milliseconds taken by all reduce tasks=81862
    Total megabyte-milliseconds taken by all map tasks=463624192
    Total megabyte-milliseconds taken by all reduce tasks=83826688
  Map-Reduce Framework
    Map input records=20033647
    Map output records=29027042
    Map output bytes=801907084
    Map output materialized bytes=919961372
    Input split bytes=3944
    Combine input records=0
    Combine output records=0
    Reduce input groups=70
    Reduce shuffle bytes=919961372

```

```

Administrator: Command Prompt
  Spilled Records=58054084
  Shuffled Maps =34
  Failed Shuffles=0
  Merged Map outputs=34
  GC time elapsed (ms)=9770
  CPU time spent (ms)=168814
  Physical memory (bytes) snapshot=16091676672
  Virtual memory (bytes) snapshot=24966234112
  Total committed heap usage (bytes)=18936233984
  Peak Map Physical memory (bytes)=803291904
  Peak Map Virtual memory (bytes)=770179072
  Peak Reduce Physical memory (bytes)=891060224
  Peak Reduce Virtual memory (bytes)=968486912
  Shuffle Errors
    BAD_ID=0
    CONNECTION=0
    IO_ERROR=0
    WRONG_LENGTH=0
    WRONG_MAP=0
    WRONG_REDUCE=0
  File Input Format Counters
    Bytes Read=4459564440
  File Output Format Counters
    Bytes Written=2778

C:\mutual_funds\java>hdfs dfs -cat /output/avgnav/part-r-00000
ABN AMRO Mutual Fund      8.495423276450284
Aditya Birla Sun Life Mutual Fund      51.54632358905095
Axis Mutual Fund          252.67322171603334
BNP Paribas Mutual Fund 150.833996634451
BOI AXA Mutual Fund       370.7302823652751
Rajaj Finance Mutual Fund 1001.588126666667
Bandhan Mutual Fund       101.70597706152809
Bank of India Mutual Fund 309.09007003910153
Baroda BNP Paribas Mutual Fund 407.67449228260926
Baroda Mutual Fund        335.19318625425757
Baroda Pioneer Mutual Fund 70.86952338264905
Benchmark Mutual Fund     539.9930477407532
Canara Robeco Mutual Fund 304.3134889822302
DBS Chola Mutual Fund     13.046567809277086
DHFL Pramerica Mutual Fund 285.3792996988044
DSP Mutual Fund 176.1134082033183
Daiwa Mutual Fund         876.4580156144862
Deutsche Mutual Fund      19.642051254233994
Edelweiss Mutual Fund     411.9452172342718
Essel Mutual Fund         45.96607933383853
Fidelity Mutual Fund      11.54057890425493
Fortis Mutual Fund        7.3791963834660305
Franklin Templeton Mutual Fund 213.2015120081551

```

```
Administrator: Command Prompt
Goldman Sachs Mutual Fund      454.44041497468135
Groww Mutual Fund             561.344953543367
IDFC Mutual Fund              67.47391863118045
KSRIC Mutual Fund             142.43998034015482
ICICI Prudential Mutual Fund  166.92976390851885
IDBI Mutual Fund              313.07375068411557
IDFC Mutual Fund              282.995426948463
IIFL Mutual Fund              117.83881441474908
IMC Mutual Fund 12.87625921190372
ITI Mutual Fund 400.756555273818
Indiabulls Mutual Fund        705.5583149272842
Invesco Mutual Fund           573.7184546190725
JM Financial Mutual Fund       35.28285629550913
JP Morgan Mutual Fund         11.827511848538387
Kotak Mahindra Mutual Fund    144.1510489886881
L&T Mutual Fund 138.1132390974859
LIC Mutual Fund 105.27472434497346
Mahindra Manulife Mutual Fund  414.23907400033255
Mirae Asset Mutual Fund 463.6301709878512
Morgan Stanley Mutual Fund    232.4392083949698
Motilal Oswal Mutual Fund     39.40763005644218
NJ Mutual Fund 203.5112880725466
Navi Mutual Fund              306.1197909612337
Nippon India Mutual Fund      191.7623348586456
PCPI India Mutual Fund        320.26439493537885
PPFAS Mutual Fund             581.6381416075076
PRINCIPAL Mutual Fund         70.45760881931315
PineBridge Mutual Fund        731.3427952337898
Principal Mutual Fund         340.6634441801029
Quant Mutual Fund             2.3769566038729555
Quantum Mutual Fund           125.82954324059918
Reliance Mutual Fund          57.140508240708315
SBI Mutual Fund 205.89704066326559
Sahara Mutual Fund            408.2386597385202
Samco Mutual Fund             327.5822425068123
Shinsei Mutual Fund           772.8720308303041
Shriram Mutual Fund           13.646590021012662
Standard Chartered Mutual Fund 60.59890570326526
Sundaram Mutual Fund          48.16261286599233
Tata Mutual Fund              212.85904350484617
Taurus Mutual Fund            365.2445480461986
Trust Mutual Fund             1027.1970495444606
UTI Mutual Fund 176.7834840427418
Union Mutual Fund             405.40196245922874
WhiteOak Capital Mutual Fund   717.4385440295713
YES Mutual Fund 1001.1739751724142
quant Mutual Fund             34.64474936084248

C:\mutual_funds\java>
```

6.3 Scheme Category Count

Objective: count how many records belong to each Scheme Category.

```
import java.io.IOException;
```

```
import org.apache.hadoop.conf.Configuration;
```

```
import org.apache.hadoop.fs.Path;
```

```
import org.apache.hadoop.io.IntWritable;
```

```
import org.apache.hadoop.io.LongWritable;
```

```
import org.apache.hadoop.io.Text;
```

```
import org.apache.hadoop.mapreduce.Job;
```

```
import org.apache.hadoop.mapreduce.Mapper;
```

```
import org.apache.hadoop.mapreduce.Reducer;
```

```
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
```

```
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
```

```
public class CategoryCount {
```

```

public static class CategoryMapper
    extends Mapper<LongWritable, Text, Text, IntWritable> {

    private static final IntWritable ONE = new IntWritable(1);
    private final Text outKey = new Text();

    @Override
    public void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException {

        String line = value.toString().trim();

        if (line.isEmpty() || line.startsWith("Fund_House,")) {
            return;
        }

        String[] fields = line.split(",", -1);

        if (fields.length < 7) {
            return;
        }

        String category = fields[2].trim();

        if (!category.isEmpty()) {

```

```

        outKey.set(category);
        context.write(outKey, ONE);
    }
}
}

```

```

public static class CategoryReducer
    extends Reducer<Text, IntWritable, Text, IntWritable> {

    @Override
    public void reduce(Text key, Iterable<IntWritable> values,
        Context context)
        throws IOException, InterruptedException {

        int total = 0;

        for (IntWritable value : values) {
            total += value.get();
        }

        context.write(key, new IntWritable(total));
    }
}

```

```

public static void main(String[] args) throws Exception {

```

```

if (args.length != 2) {
    System.err.println(
        "Usage: CategoryCount <input path> <output path>");
    System.exit(2);
}

Configuration conf = new Configuration();
Job job = Job.getInstance(conf, "Scheme Category Count");

job.setJarByClass(CategoryCount.class);
job.setMapperClass(CategoryMapper.class);
job.setReducerClass(CategoryReducer.class);

job.setMapOutputKeyClass(Text.class);
job.setMapOutputValueClass(IntWritable.class);
job.setOutputKeyClass(Text.class);
job.setOutputValueClass(IntWritable.class);

job.setNumReduceTasks(5);

FileInputFormat.addInputPath(job, new Path(args[0]));
FileOutputFormat.setOutputPath(job, new Path(args[1]));

System.exit(job.waitForCompletion(true) ? 0 : 1);
}
}

```

The Mapper emits Scheme_Category → 1 and the Reducer sums the values for each category.

```
Administrator: Command Prompt
C:\mutual_funds\java>notepad CategoryCount.java

C:\mutual_funds\java>javac -classpath "%HADOOP_HOME%\share\hadoop\common\*;%HADOOP_HOME%\share\hadoop\common\lib\*;%HADOOP_HOME%\share\hadoop\mapreduce\*;%HADOOP_HOME%\share\hadoop\mapreduce\lib\*" CategoryCount.java

C:\mutual_funds\java>jar -cvf CategoryCount.jar *.class
added manifest
adding: AverageMAV$AvgMapper.class(in = 1915) (out= 794)(deflated 58%)
adding: AverageMAV$AvgReducer.class(in = 1698) (out= 727)(deflated 57%)
adding: AverageMAV.class(in = 1464) (out= 886)(deflated 44%)
adding: CategoryCount$CategoryMapper.class(in = 1923) (out= 799)(deflated 58%)
adding: CategoryCount$CategoryReducer.class(in = 1657) (out= 691)(deflated 58%)
adding: CategoryCount.class(in = 1489) (out= 799)(deflated 46%)
adding: MaxMAV$MaxMapper.class(in = 1922) (out= 788)(deflated 58%)
adding: MaxMAV$MaxReducer.class(in = 1749) (out= 747)(deflated 57%)
adding: MaxMAV.class(in = 1434) (out= 793)(deflated 44%)

C:\mutual_funds\java>hdfs dfs -rm -r /output/categorycount
rm: /output/categorycount: No such file or directory

C:\mutual_funds\java>hadoop jar CategoryCount.jar CategoryCount /mutual_funds/Mutual_Funds.csv /output/categorycount
2026-09-11 18:44:39,481 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 18:44:39,971 INFO mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-11 18:44:40,177 INFO input.FileInputFormat: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/ASUS/.staging/job_1789129568662_0003
2026-09-11 18:44:40,246 INFO mapreduce.JobSubmitter: Total input files to process : 1
2026-09-11 18:44:40,327 INFO mapreduce.JobSubmitter: number of splits:34
2026-09-11 18:44:40,329 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789129568662_0003
2026-09-11 18:44:40,329 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-11 18:44:40,435 INFO conf.Configuration: resource-types.xml not found
2026-09-11 18:44:40,435 INFO resource.ResourceUtil: Unable to find 'resource-types.xml'.
2026-09-11 18:44:40,500 INFO impl.YarnClientImpl: Submitted application application_1789129568662_0003
2026-09-11 18:44:40,525 INFO mapreduce.Job: The url to track the job: http://MDU:8088/proxy/application_1789129568662_0003/
2026-09-11 18:44:40,526 INFO mapreduce.Job: Running job: job_1789129568662_0003
2026-09-11 18:44:47,699 INFO mapreduce.Job: Job job_1789129568662_0003 running in uber mode : false
2026-09-11 18:44:47,701 INFO mapreduce.Job: map 0% reduce 0%
2026-09-11 18:45:06,786 INFO mapreduce.Job: map 3% reduce 0%
2026-09-11 18:45:07,832 INFO mapreduce.Job: map 18% reduce 0%
2026-09-11 18:45:22,473 INFO mapreduce.Job: map 24% reduce 0%
2026-09-11 18:45:23,502 INFO mapreduce.Job: map 32% reduce 0%
2026-09-11 18:45:24,704 INFO mapreduce.Job: map 35% reduce 0%
2026-09-11 18:45:35,867 INFO mapreduce.Job: map 44% reduce 0%
2026-09-11 18:45:36,890 INFO mapreduce.Job: map 47% reduce 0%
2026-09-11 18:45:39,996 INFO mapreduce.Job: map 47% reduce 16%
2026-09-11 18:45:41,039 INFO mapreduce.Job: map 50% reduce 16%
2026-09-11 18:45:46,138 INFO mapreduce.Job: map 59% reduce 17%
2026-09-11 18:45:51,262 INFO mapreduce.Job: map 62% reduce 17%
2026-09-11 18:45:52,302 INFO mapreduce.Job: map 62% reduce 21%
2026-09-11 18:45:55,377 INFO mapreduce.Job: map 65% reduce 21%
```

```
Administrator: Command Prompt
2026-09-11 18:45:52,302 INFO mapreduce.Job: map 62% reduce 21%
2026-09-11 18:45:55,377 INFO mapreduce.Job: map 65% reduce 21%
2026-09-11 18:45:56,428 INFO mapreduce.Job: map 68% reduce 21%
2026-09-11 18:45:57,459 INFO mapreduce.Job: map 74% reduce 21%
2026-09-11 18:45:58,529 INFO mapreduce.Job: map 74% reduce 25%
2026-09-11 18:46:03,683 INFO mapreduce.Job: map 76% reduce 25%
2026-09-11 18:46:07,795 INFO mapreduce.Job: map 79% reduce 25%
2026-09-11 18:46:08,833 INFO mapreduce.Job: map 88% reduce 25%
2026-09-11 18:46:10,904 INFO mapreduce.Job: map 88% reduce 29%
2026-09-11 18:46:13,967 INFO mapreduce.Job: map 91% reduce 29%
2026-09-11 18:46:16,001 INFO mapreduce.Job: map 94% reduce 31%
2026-09-11 18:46:17,020 INFO mapreduce.Job: map 100% reduce 31%
2026-09-11 18:46:22,083 INFO mapreduce.Job: map 100% reduce 51%
2026-09-11 18:46:28,139 INFO mapreduce.Job: map 100% reduce 80%
2026-09-11 18:46:34,199 INFO mapreduce.Job: map 100% reduce 93%
2026-09-11 18:46:39,264 INFO mapreduce.Job: map 100% reduce 100%
2026-09-11 18:46:39,272 INFO mapreduce.Job: Job job_1789129568662_0003 completed successfully
2026-09-11 18:46:39,450 INFO mapreduce.Job: Counters: 55

File System Counters
  FILE: Number of bytes read=779667640
  FILE: Number of bytes written=1567690425
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=4459568384
  HDFS: Number of bytes written=2422
  HDFS: Number of read operations=107
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0

Job Counters
  Killed map tasks=1
  Launched map tasks=35
  Launched reduce tasks=1
  Data-local map tasks=35
  Total time spent by all maps in occupied slots (ms)=431382
  Total time spent by all reduces in occupied slots (ms)=75776
  Total time spent by all map tasks (ms)=431382
  Total time spent by all reduce tasks (ms)=75776
  Total vcore-milliseconds taken by all map tasks=431382
  Total vcore-milliseconds taken by all reduce tasks=75776
  Total megabyte-milliseconds taken by all map tasks=441735168
  Total megabyte-milliseconds taken by all reduce tasks=77594624

Map-Reduce Framework
  Map input records=29033647
  Map output records=29033646
  Map output bytes=721600330
  Map output materialized bytes=779667826
  Input split bytes=3944
```

```
Administrator: Command Prompt

Input split bytes=3944
Combine input records=0
Combine output records=0
Reduce input groups=83
Reduce shuffle bytes=779667826
Reduce input records=29033646
Reduce output records=83
Spilled Records=58067292
Shuffled Maps=34
Failed Shuffles=0
Merged Map outputs=34
GC time elapsed (ms)=9384
CPU time spent (ms)=161055
Physical memory (bytes) snapshot=15986319360
Virtual memory (bytes) snapshot=24265617888
Total committed heap usage (bytes)=18318622720
Peak Map Physical memory (bytes)=501149696
Peak Map Virtual memory (bytes)=742989824
Peak Reduce Physical memory (bytes)=898093056
Peak Reduce Virtual memory (bytes)=946397184

Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0

File Input Format Counters
  Bytes Read=4459564440
File Output Format Counters
  Bytes Written=2422

C:\mutual_funds\java>hdfs dfs -cat /output/categorycount/part-r-00000
1          31464
1098 Days      18101
1099 Days      399333
1100 DAYS       2
1100 Days      9948
1100 days      30443
1102 Days      2924
1111 DAYS       15
1116 Days      10108
1124 Days      6744
1141 Days      8718
1150 DAYS       888
1154 DAYS      26340
1305 Days      1468
19 months Plan 39684
466 DAYS       782
```

```
Administrator: Command Prompt

5 Year Plan      44052
54EB Growth      14822
91 Days 3
Analyst's Conviction Equalized 7318
Annual Dividend 339
Balanced         1918
Compulsory Reinvestment 1522
DIRECT 56540
Daily 3988
Debt Scheme - Banking and PSU Fund 402167
Debt Scheme - Corporate Bond Fund 454097
Debt Scheme - Credit Risk Fund 360615
Debt Scheme - Dynamic Bond 210640
Debt Scheme - Floater Fund 240331
Debt Scheme - Gilt Fund 416007
Debt Scheme - Gilt Fund with 10 year constant duration 78945
Debt Scheme - Liquid Fund 374998
Debt Scheme - Long Duration Fund 52954
Debt Scheme - Low Duration Fund 904107
Debt Scheme - Medium Duration Fund 381734
Debt Scheme - Medium to Long Duration Fund 330954
Debt Scheme - Money Market Fund 502398
Debt Scheme - Overnight Fund 341745
Debt Scheme - Short Duration Fund 684265
Debt Scheme - Ultra Short Duration Fund 641459
Direct 140356
ELSS 164729
Equity Scheme - Contra Fund 40585
Equity Scheme - Dividend Yield Fund 74349
Equity Scheme - ELSS 80740
Equity Scheme - Focused Fund 36044
Equity Scheme - Focussed Fund 2296
Equity Scheme - Large & Mid Cap Fund 346229
Equity Scheme - Large Cap Fund 410037
Equity Scheme - Mid Cap Fund 296486
Equity Scheme - Multi Cap Fund 172019
Equity Scheme - Sectoral/ Thematic 36990
Equity Scheme - Small Cap Fund 224544
Equity Scheme - Value Fund 184200
FV Rs 32.161 9917
Formerly Known as IIFL Mutual Fund 3532025
Formerly Super Institutional Plan 13103
G 3570
Gilt 18601
Growth 538864
Half Yearly Dividend 47684
Hybrid Scheme - Aggressive Hybrid Fund 423581
Hybrid Scheme - Arbitrage Fund 350647
Hybrid Scheme - Balanced Hybrid Fund 32205
```

```

Administrator: Command Prompt
Hybrid Scheme - Conservative Hybrid Fund      418202
Hybrid Scheme - Dynamic Asset Allocation or Balanced Advantage  206987
Hybrid Scheme - Equity Savings  297990
Hybrid Scheme - Multi Asset Allocation  130525
IDF      2905329
Income  8991054
Liquid  65815
Merger of Capex & Energy Opportunities  8450
Money Market  50510
Other Scheme - FoF Domestic      504240
Other Scheme - FoF Overseas      273042
Other Scheme - Gold ETF  35364
Other Scheme - Index Funds      436441
Other Scheme - Other  ETFs      168205
Payout  41759
Solution Oriented Scheme - Children's Fund      75413
Solution Oriented Scheme - Retirement Fund      135332
erstwhile Cash Option  6275

C:\mutual_funds\java>

```

7. Apache Pig Analysis

Apache Pig provides a higher-level data-flow language for working with large datasets. In this project, Pig Latin is used to load the HDFS CSV, define schema fields, filter records, group records, and apply aggregate functions such as AVG, MAX, and COUNT.

All five Pig experiments were executed from the project directory using Apache Pig 0.18.0 in MapReduce mode.

```

Administrator: Command Prompt

:\mutual_funds\java>mkdir C:\mutual_funds\pig

:\mutual_funds\java>cd C:\mutual_funds\pig

:\mutual_funds\pig>pig -version
Apache Pig version 0.18.0 (r1926484)
Compiled Jun 16 2025, 10:38:21

:\mutual_funds\pig>

```

7.1 Load and Display Mutual Fund Data

Objective: load the CSV dataset using PigStorage and display the first ten records. A separate alias name, `sample_data`, is used for the LIMIT result.

```

funds = LOAD '/mutual_funds/Mutual_Funds.csv'
USING PigStorage(',')
AS (
Fund_House:chararray,
Scheme_Type:chararray,
Scheme_Category:chararray,

```

Scheme_Code:chararray,
Scheme_Name:chararray,
Date:chararray,
NAV:double
);

sample_data = LIMIT funds 10;
DUMP sample_data;

C:\mutual_funds\pig>notepad pig1_load.pig

```
Administrator: Command Prompt
C:\mutual_funds\pig>pig pig1_load.pig
2026-09-11 18:55:25,455 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-11 18:55:25,457 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-11 18:55:25,458 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-11 18:55:25,508 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-11 18:55:25,594 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig-1789131325590.log
2026-09-11 18:55:25,878 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\VASUS\.pigbootstrap not found
2026-09-11 18:55:25,908 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-11 18:55:25,908 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-11 18:55:26,179 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: pig-pig1_load.pig-09c974d8-599c-4bc8-8e96-33b4b7b94e27
2026-09-11 18:55:26,179 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-11 18:55:26,594 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: LIMIT
2026-09-11 18:55:26,613 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-11 18:55:26,644 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-11 18:55:26,697 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-11 18:55:26,744 [main] INFO org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter - File Output Committer Algorithm version is 2
2026-09-11 18:55:26,744 [main] INFO org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter - FileOutputCommitter skip cleanup_temporary folders under output directory:false, ignore cleanup failures: false
2026-09-11 18:55:26,794 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-11 18:55:26,836 [main] WARN org.apache.pig.data.SchemaTupleBackend - SchemaTupleBackend has already been initialized
2026-09-11 18:55:26,836 [main] INFO org.apache.pig.builtin.PigStorage - Using PigTextInputFormat
2026-09-11 18:55:26,836 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-11 18:55:26,844 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
2026-09-11 18:55:26,879 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.task.id is deprecated. Instead, use mapreduce.task.attempt.id
2026-09-11 18:55:27,012 [main] INFO org.apache.hadoop.mapreduce.lib.output.FileOutputCommitter - Saved output of task 'attempt_0001_m_000001_1' to hdfs://localhost:9000/tmp/temp-913129602/tmp1366548521
2026-09-11 18:55:27,024 [main] WARN org.apache.pig.data.SchemaTupleBackend - SchemaTupleBackend has already been initialized
2026-09-11 18:55:27,035 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-11 18:55:27,035 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
(Fund_House, Scheme_Type, Scheme_Category, Scheme_Name, Date,)
(Standard Chartered Mutual Fund, Open Ended Schemes, Income, 100027, Grindlays Super Saver Income Fund-GSSIF-Half Yearly Dividend, 29-05-2008, 10.7205)
(Standard Chartered Mutual Fund, Open Ended Schemes, Income, 100027, Grindlays Super Saver Income Fund-GSSIF-Half Yearly Dividend, 28-05-2008, 10.725)
(Standard Chartered Mutual Fund, Open Ended Schemes, Income, 100027, Grindlays Super Saver Income Fund-GSSIF-Half Yearly Dividend, 27-05-2008, 10.7216)
(Standard Chartered Mutual Fund, Open Ended Schemes, Income, 100027, Grindlays Super Saver Income Fund-GSSIF-Half Yearly Dividend, 26-05-2008, 10.7206)
(Standard Chartered Mutual Fund, Open Ended Schemes, Income, 100027, Grindlays Super Saver Income Fund-GSSIF-Half Yearly Dividend, 23-05-2008, 10.7152)
(Standard Chartered Mutual Fund, Open Ended Schemes, Income, 100027, Grindlays Super Saver Income Fund-GSSIF-Half Yearly Dividend, 22-05-2008, 10.7136)
(Standard Chartered Mutual Fund, Open Ended Schemes, Income, 100027, Grindlays Super Saver Income Fund-GSSIF-Half Yearly Dividend, 21-05-2008, 10.721)
(Standard Chartered Mutual Fund, Open Ended Schemes, Income, 100027, Grindlays Super Saver Income Fund-GSSIF-Half Yearly Dividend, 20-05-2008, 10.7202)
(Standard Chartered Mutual Fund, Open Ended Schemes, Income, 100027, Grindlays Super Saver Income Fund-GSSIF-Half Yearly Dividend, 16-05-2008, 10.7134)
2026-09-11 18:55:27,104 [main] INFO org.apache.pig.Main - Pig script completed in 1 second and 671 milliseconds (1671 ms)
C:\mutual_funds\pig>
```

Observed execution: the job successfully read 29,033,647 records (4,459,577,292 bytes) from HDFS and displayed sample records.

7.2 NAV Greater Than 1000

Objective: filter the dataset so that only records with NAV greater than 1000 are retained, then display the first 20 matching records.

```
funds = LOAD '/mutual_funds/Mutual_Funds.csv'
USING PigStorage(',')
AS (
Fund_House:chararray,
Scheme_Type:chararray,
Scheme_Category:chararray,
Scheme_Code:chararray,
```

Scheme_Name:chararray,

Date:chararray,

NAV:double

);

high_nav = FILTER funds BY NAV > 1000;

result = LIMIT high_nav 20;

DUMP result;

```
Administrator: Command Prompt
C:\hadoop\sbin>cd C:\mutual_funds\pig

C:\mutual_funds\pig>pig pig2_highnav.pig
2026-09-11 19:10:04,787 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-11 19:10:04,788 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-11 19:10:04,788 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-11 19:10:04,922 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-11 19:10:04,922 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1789132204919.log
2026-09-11 19:10:05,185 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\User-VAGUS\pigbootstrap not found
2026-09-11 19:10:05,217 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-11 19:10:05,217 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-11 19:10:05,491 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-pig2_highnav.pig-04313b3a-4234-4460-bbb4-a953c72a60b2
2026-09-11 19:10:05,491 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-11 19:10:05,906 [main] WARN org.apache.pig.newplan.BaseOperatorPlan - Encountered Warning IMPLICIT_CAST_TO_DOUBLE 1 time(s).
2026-09-11 19:10:05,912 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: FILTER,LIMIT
2026-09-11 19:10:05,927 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-11 19:10:05,948 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - (RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByCon
stParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantC
alculator, SplitFilter, StreamTypeCastInserter])
2026-09-11 19:10:05,968 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 6994800192 to monitor, collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-11 19:10:06,018 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-11 19:10:06,035 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 2
2026-09-11 19:10:06,035 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 2
2026-09-11 19:10:06,076 [main] INFO org.apache.hadoop.yarn.client.NMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:10:06,201 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-11 19:10:06,211 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-11 19:10:06,215 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-11 19:10:06,215 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-11 19:10:06,217 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-11 19:10:06,219 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers.
2026-09-11 19:10:06,219 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting Parallelism to 1
2026-09-11 19:10:06,220 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.reduce.tasks is deprecated. Instead, use mapreduce.job.reduces
2026-09-11 19:10:06,220 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - This job cannot be converted run in-process
2026-09-11 19:10:06,230 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.submit.replication is deprecated. Instead, use mapreduce.client.submit.file.replication
2026-09-11 19:10:06,356 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/pig-0.18.0-core-h3.jar to DistributedCache through /tmp/temp-54230878
1/tmp-351114548/pig-0.18.0-core-h3.jar
2026-09-11 19:10:06,384 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/lib/automaton-1.11-8.jar to DistributedCache through /tmp/temp-542308
781/tmp683317955/automaton-1.11-8.jar
2026-09-11 19:10:06,411 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/lib/antlr-runtime-3.4.jar to DistributedCache through /tmp/temp-54230
8781/tmp702170826/antlr-runtime-3.4.jar
2026-09-11 19:10:06,437 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/lib/joda-time-2.14.0.jar to DistributedCache through /tmp/temp-542308
781/tmp-367414888/joda-time-2.14.0.jar
2026-09-11 19:10:06,474 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/lib/RoaringBitmap-shaded-0.7.14.jar to DistributedCache through /tmp/
temp-542308781/tmp1665197476/RoaringBitmap-shaded-0.7.14.jar
2026-09-11 19:10:06,508 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting up single store job
2026-09-11 19:10:06,512 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Key [pig.schematuple] is false, will not generate code.
2026-09-11 19:10:06,512 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Starting process to move generated code to distributed cache
2026-09-11 19:10:06,512 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Setting key [pig.schematuple.classes] with classes to deserialize []
2026-09-11 19:10:06,527 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
```

```

Administrator: Command Prompt
2026-09-11 19:10:06,553 [JobControl] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:10:06,562 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.task.id is deprecated. Instead, use mapreduce.task.attempt.id
2026-09-11 19:10:06,780 [JobControl] INFO org.apache.hadoop.mapreduce.JobResourceUploader - Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/ASUS/staging/job_1789129568662_0006
2026-09-11 19:10:06,787 [JobControl] WARN org.apache.hadoop.mapreduce.JobResourceUploader - No job jar file set. User classes may not be found. See Job or Job#setJar(String).
2026-09-11 19:10:06,814 [JobControl] INFO org.apache.pig.builtin.PigStorage - Using PigTextInputFormat
2026-09-11 19:10:06,817 [JobControl] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-11 19:10:06,817 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
2026-09-11 19:10:06,830 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths (combined) to process : 34
2026-09-11 19:10:06,851 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - number of splits:34
2026-09-11 19:10:06,964 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-11 19:10:07,022 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Submitting tokens for job: job_1789129568662_0006
2026-09-11 19:10:07,023 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Executing with tokens: []
2026-09-11 19:10:07,182 [JobControl] INFO org.apache.hadoop.mapred.YARNRunner - Job jar is not present. Not adding any jar to the list of resources.
2026-09-11 19:10:07,132 [JobControl] INFO org.apache.hadoop.conf.Configuration - resource-types.xml not found
2026-09-11 19:10:07,134 [JobControl] INFO org.apache.hadoop.yarn.util.resource.ResourceUtils - Unable to find 'resource-types.xml'.
2026-09-11 19:10:07,177 [JobControl] INFO org.apache.hadoop.yarn.client.api.impl.YarnClientImpl - Submitted application application_1789129568662_0006
2026-09-11 19:10:07,201 [JobControl] INFO org.apache.hadoop.mapreduce.Job - The url to track the job: http://H0U:8088/proxy/application_1789129568662_0006/
2026-09-11 19:10:07,201 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - HadoopJobId: job_1789129568662_0006
2026-09-11 19:10:07,201 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Processing aliases funds,high_nav,result
2026-09-11 19:10:07,201 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - detailed locations: M: funds[1,8],funds[-1,-1],high_nav[13,11],result[15,9] C: R:
2026-09-11 19:10:07,213 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 0% complete
2026-09-11 19:10:07,213 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0006]
2026-09-11 19:10:29,864 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 4% complete
2026-09-11 19:10:29,864 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0006]
2026-09-11 19:10:44,689 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 8% complete
2026-09-11 19:10:44,697 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0006]
2026-09-11 19:10:59,444 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 16% complete
2026-09-11 19:10:59,444 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0006]
2026-09-11 19:11:09,641 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 20% complete
2026-09-11 19:11:09,641 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0006]
2026-09-11 19:11:21,993 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 25% complete
2026-09-11 19:11:22,002 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0006]
2026-09-11 19:11:34,920 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 30% complete
2026-09-11 19:11:34,922 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0006]
2026-09-11 19:11:40,020 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 50% complete
2026-09-11 19:11:40,021 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0006]
2026-09-11 19:11:42,590 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:11:42,602 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:11:43,064 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:11:43,073 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:11:43,118 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:11:43,124 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:11:43,199 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScripStats - Pig script settings are added to the job
2026-09-11 19:11:43,200 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-11 19:11:43,201 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceCompiler - Reduce phase detected, estimating # of required reducers.
2026-09-11 19:11:43,202 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceCompiler - Setting Parallelism to 1
2026-09-11 19:11:43,202 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceCompiler - This job cannot be converted run in-process
2026-09-11 19:11:43,284 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceCompiler - Added jar file:/C:/pig/pig-0.18.0-core-h3.jar to DistributedCache through /tmp/temp-54230878

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Administrator: Command Prompt
8781/tmp-484710326/jnlr-runtime-3.4.jar
2026-09-11 19:11:43,380 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceCompiler - Added jar file:/C:/pig/lib/joda-time-2.14.0.jar to DistributedCache through /tmp/temp-542308781
781/tmp597529793/joda-time-2.14.0.jar
2026-09-11 19:11:43,410 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceCompiler - Added jar file:/C:/pig/lib/RoaringBitmap-shaded-0.7.14.jar to DistributedCache through /tmp/temp-542308781
tmp405777609/RoaringBitmap-shaded-0.7.14.jar
2026-09-11 19:11:43,432 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceCompiler - Setting up single store job
2026-09-11 19:11:43,434 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Key [pig.schematuple] is false, will not generate code.
2026-09-11 19:11:43,434 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Starting process to move generated code to distributed cache
2026-09-11 19:11:43,435 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Setting key [pig.schematuple.classes] with classes to deserialize []
2026-09-11 19:11:43,455 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
2026-09-11 19:11:43,461 [JobControl] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:11:43,467 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-11 19:11:43,482 [JobControl] INFO org.apache.hadoop.mapreduce.JobResourceUploader - Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/ASUS/staging/job_1789129568662_0007
2026-09-11 19:11:43,485 [JobControl] WARN org.apache.hadoop.mapreduce.JobResourceUploader - No job jar file set. User classes may not be found. See Job or Job#setJar(String).
2026-09-11 19:11:43,524 [JobControl] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-11 19:11:43,524 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
2026-09-11 19:11:43,585 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - number of splits:1
2026-09-11 19:11:43,681 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Submitting tokens for job: job_1789129568662_0007
2026-09-11 19:11:43,681 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Executing with tokens: []
2026-09-11 19:11:43,686 [JobControl] INFO org.apache.hadoop.mapred.YARNRunner - Job jar is not present. Not adding any jar to the list of resources.
2026-09-11 19:11:43,941 [JobControl] INFO org.apache.hadoop.yarn.client.api.impl.YarnClientImpl - Submitted application application_1789129568662_0007
2026-09-11 19:11:43,950 [JobControl] INFO org.apache.hadoop.mapreduce.Job - The url to track the job: http://H0U:8088/proxy/application_1789129568662_0007/
2026-09-11 19:11:43,969 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - HadoopJobId: job_1789129568662_0007
2026-09-11 19:11:43,969 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Processing aliases
2026-09-11 19:11:43,970 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - detailed locations: M: C: R:
2026-09-11 19:11:58,030 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 75% complete
2026-09-11 19:11:58,032 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0007]
2026-09-11 19:12:01,089 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0007]
2026-09-11 19:12:04,182 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:12:04,190 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:12:04,325 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:12:04,332 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:12:04,370 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:12:04,374 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:12:04,401 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 100% complete
2026-09-11 19:12:04,417 [main] INFO org.apache.pig.tools.pigstats.mapreduce.SimplePigStats - Script Statistics:

HadoopVersion PigVersion UserId StartedAt FinishedAt Features
3.2.4 0.18.0 ASUS 2026-09-11 19:10:06 2026-09-11 19:12:04 FILTER,LIMIT

Success!

Job Stats (time in seconds):
JobId Maps Reduces MaxMapTime MinMapTime AvgMapTime MedianMapTime MaxReduceTime MinReduceTime AvgReduceTime MedianReduceTime Alias Feature Outputs
job_1789129568662_0006 34 1 14 6 11 12 56 56 56 funds,high_nav,result
job_1789129568662_0007 1 1 2 2 2 2 2 2 2 hdfs://localhost:9000/tmp/temp-542308781/tmp-1233532388,

```

```
Administrator: Command Prompt

Input(s):
Successfully read 29033647 records (4459577292 bytes) from: "/mutual_funds/Mutual_Funds.csv"

Output(s):
Successfully stored 20 records (2980 bytes) in: "hdfs://localhost:9000/tmp/temp-542388781/tmp-1233532388"

Counters:
Total records written : 20
Total bytes written : 2980
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1789129568662_0006 -> job_1789129568662_0007,
job_1789129568662_0007

2026-09-11 19:12:04,421 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:12:04,425 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:12:04,475 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:12:04,479 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:12:04,503 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:12:04,508 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:12:04,538 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:12:04,543 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:12:04,575 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:12:04,579 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:12:04,601 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:12:04,605 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:12:04,632 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCARDED_TYPE_CONVERSION_FAILED 2 time(s).
2026-09-11 19:12:04,632 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-11 19:12:04,637 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-11 19:12:04,638 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-11 19:12:04,643 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input files to process : 1
2026-09-11 19:12:04,643 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
(N) Mutual Fund,Open Ended Schemes,Debt Scheme - Overnight Fund,150368,NJ Overnight Fund - Regular Plan - Growth Option,02-07-2023,1055.8616)
(N) Mutual Fund,Open Ended Schemes,Debt Scheme - Overnight Fund,150368,NJ Overnight Fund - Regular Plan - Growth Option,03-07-2023,1056.8358)
(N) Mutual Fund,Open Ended Schemes,Debt Scheme - Overnight Fund,150368,NJ Overnight Fund - Regular Plan - Growth Option,04-07-2023,1056.2116)
(N) Mutual Fund,Open Ended Schemes,Debt Scheme - Overnight Fund,150368,NJ Overnight Fund - Regular Plan - Growth Option,05-07-2023,1056.3885)
(N) Mutual Fund,Open Ended Schemes,Debt Scheme - Overnight Fund,150368,NJ Overnight Fund - Regular Plan - Growth Option,06-07-2023,1056.6391)
(N) Mutual Fund,Open Ended Schemes,Debt Scheme - Overnight Fund,150368,NJ Overnight Fund - Regular Plan - Growth Option,09-07-2023,1057.1707)
(N) Mutual Fund,Open Ended Schemes,Debt Scheme - Overnight Fund,150368,NJ Overnight Fund - Regular Plan - Growth Option,10-07-2023,1057.3508)
(N) Mutual Fund,Open Ended Schemes,Debt Scheme - Overnight Fund,150368,NJ Overnight Fund - Regular Plan - Growth Option,11-07-2023,1057.5304)
(N) Mutual Fund,Open Ended Schemes,Debt Scheme - Overnight Fund,150368,NJ Overnight Fund - Regular Plan - Growth Option,12-07-2023,1057.7062)
(N) Mutual Fund,Open Ended Schemes,Debt Scheme - Overnight Fund,150368,NJ Overnight Fund - Regular Plan - Growth Option,13-07-2023,1057.8806)
(N) Mutual Fund,Open Ended Schemes,Debt Scheme - Overnight Fund,150368,NJ Overnight Fund - Regular Plan - Growth Option,14-07-2023,1058.0563)
(N) Mutual Fund,Open Ended Schemes,Debt Scheme - Overnight Fund,150368,NJ Overnight Fund - Regular Plan - Growth Option,16-07-2023,1058.4093)
(N) Mutual Fund,Open Ended Schemes,Debt Scheme - Overnight Fund,150368,NJ Overnight Fund - Regular Plan - Growth Option,17-07-2023,1058.586)
(N) Mutual Fund,Open Ended Schemes,Debt Scheme - Overnight Fund,150368,NJ Overnight Fund - Regular Plan - Growth Option,18-07-2023,1058.7629)
(N) Mutual Fund,Open Ended Schemes,Debt Scheme - Overnight Fund,150368,NJ Overnight Fund - Regular Plan - Growth Option,19-07-2023,1058.939)
(N) Mutual Fund,Open Ended Schemes,Debt Scheme - Overnight Fund,150368,NJ Overnight Fund - Regular Plan - Growth Option,20-07-2023,1059.1211)
(N) Mutual Fund,Open Ended Schemes,Debt Scheme - Overnight Fund,150368,NJ Overnight Fund - Regular Plan - Growth Option,23-07-2023,1059.6728)
(N) Mutual Fund,Open Ended Schemes,Debt Scheme - Overnight Fund,150368,NJ Overnight Fund - Regular Plan - Growth Option,27-06-2023,1054.9175)
(N) Mutual Fund,Open Ended Schemes,Debt Scheme - Overnight Fund,150368,NJ Overnight Fund - Regular Plan - Growth Option,29-06-2023,1055.2948)
(N) Mutual Fund,Open Ended Schemes,Debt Scheme - Overnight Fund,150368,NJ Overnight Fund - Regular Plan - Growth Option,30-06-2023,1055.4833)
2026-09-11 19:12:04,858 [main] INFO org.apache.pig.Main - Pig script completed in 2 minutes and 59 milliseconds (120059 ms)

C:\mutual_funds>pig>
```

The result demonstrates that Pig can apply a numeric predicate directly to the NAV field and pass only qualifying records to the LIMIT operation.

7.3 Average NAV by Scheme Category

Objective: group records by Scheme_Category and calculate the average NAV in each category.

```
funds = LOAD '/mutual_funds/Mutual_Funds.csv'
USING PigStorage(',')
AS (
Fund_House:chararray,
Scheme_Type:chararray,
Scheme_Category:chararray,
Scheme_Code:chararray,
Scheme_Name:chararray,
```

Date:chararray,

NAV:double

);

grouped_data = GROUP funds BY Scheme_Category;

avg_nav = FOREACH grouped_data

GENERATE

group AS Scheme_Category,

AVG(funds.NAV) AS Average_NAV;

DUMP avg_nav;

```
Administrator: Command Prompt
C:\mutual_funds\pig>notepad pig3_avgcategory.pig

C:\mutual_funds\pig>pig pig3_avgcategory.pig
2026-09-11 19:14:47,980 [INFO] pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-11 19:14:47,981 [INFO] pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-11 19:14:47,981 [INFO] pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-11 19:14:48,055 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-11 19:14:48,055 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig-1789132488053.log
2026-09-11 19:14:48,335 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\ASUS/.pigbootstrap not found
2026-09-11 19:14:48,359 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.job.tracker.address
2026-09-11 19:14:48,359 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-11 19:14:48,635 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-pig3_avgcategory.pig-716c5afa-7755-49a1-bf24-6f48b15a5082
2026-09-11 19:14:48,635 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline.service.enabled set to false
2026-09-11 19:14:49,059 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY
2026-09-11 19:14:49,073 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schema.tuple] was not set... will not generate code.
2026-09-11 19:14:49,090 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-11 19:14:49,120 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699480192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-11 19:14:49,145 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-11 19:14:49,156 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-11 19:14:49,173 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2026-09-11 19:14:49,173 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2026-09-11 19:14:49,216 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:14:49,337 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-11 19:14:49,345 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRS-pigState - Pig script settings are added to the job
2026-09-11 19:14:49,349 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-11 19:14:49,349 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-11 19:14:49,352 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-11 19:14:49,354 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers.
2026-09-11 19:14:49,355 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Using reducer estimator: org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator
2026-09-11 19:14:49,360 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator - BytesPerReducer=1000000000 maxReducers=999 totalInputFileSize=4459429272
2026-09-11 19:14:49,360 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting Parallelism to 5
2026-09-11 19:14:49,360 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.reduce.tasks is deprecated. Instead, use mapreduce.job.reduces
2026-09-11 19:14:49,360 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - This job cannot be converted run in-process
2026-09-11 19:14:49,372 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.submit.replication is deprecated. Instead, use mapreduce.client.submit.file.replication
2026-09-11 19:14:49,518 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/pig-0.18.0-core-h3.jar to DistributedCache through /tmp/tmp-1429656159/tmp1401167682/pig-0.18.0-core-h3.jar
2026-09-11 19:14:49,542 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/lib/automaton-1.11-8.jar to DistributedCache through /tmp/tmp-1429656159/tmp1563738368/automaton-1.11-8.jar
2026-09-11 19:14:49,569 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/lib/antlr-runtime-3.4.jar to DistributedCache through /tmp/tmp-1429656159/tmp1378672418/antlr-runtime-3.4.jar
2026-09-11 19:14:49,600 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/lib/joda-time-2.14.0.jar to DistributedCache through /tmp/tmp-1429656159/tmp-1527845125/joda-time-2.14.0.jar
2026-09-11 19:14:49,630 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/lib/RoaringBitmap-shaded-0.7.14.jar to DistributedCache through /tmp/tmp-1429656159/tmp-1953781948/RoaringBitmap-shaded-0.7.14.jar
2026-09-11 19:14:49,664 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting up single store job
```

```

Administrator: Command Prompt
2026-09-11 19:14:49,710 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
2026-09-11 19:14:49,714 [JobControl] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:14:49,726 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.task.id is deprecated. Instead, use mapreduce.task.attempt.id
2026-09-11 19:14:49,935 [JobControl] INFO org.apache.hadoop.mapreduce.JobResourceUploader - Disabling Feature Coding for path: /tmp/hadoop-yarn/staging/ASUS/staging/job_1789129568662_0008
2026-09-11 19:14:49,941 [JobControl] WARN org.apache.hadoop.mapreduce.JobResourceUploader - No job jar file set. User classes may not be found. See Job or Job#setJar(String).
2026-09-11 19:14:49,962 [JobControl] INFO org.apache.pig.builtin.PigStorage - Using PigTextInputFormat
2026-09-11 19:14:49,965 [JobControl] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-11 19:14:49,965 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
2026-09-11 19:14:49,977 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths (combined) to process : 34
2026-09-11 19:14:50,075 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - number of splits:34
2026-09-11 19:14:50,101 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-11 19:14:50,180 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Submitting tokens for job: job_1789129568662_0008
2026-09-11 19:14:50,180 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Executing with tokens: []
2026-09-11 19:14:50,260 [JobControl] INFO org.apache.hadoop.mapred.YARNRunner - Job jar is not present. Not adding any jar to the list of resources.
2026-09-11 19:14:50,302 [JobControl] INFO org.apache.hadoop.conf.Configuration - resource-types.xml not found
2026-09-11 19:14:50,302 [JobControl] INFO org.apache.hadoop.yarn.util.resource.ResourceUtils - Unable to find 'resource-types.xml'.
2026-09-11 19:14:50,340 [JobControl] INFO org.apache.hadoop.yarn.client.api.impl.YarnClientImpl - Submitted application application_1789129568662_0008
2026-09-11 19:14:50,363 [JobControl] INFO org.apache.hadoop.mapreduce.Job - The url to track the job: http://MDU:8088/proxy/application_1789129568662_0008/
2026-09-11 19:14:50,365 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - HadoopJobId: job_1789129568662_0008
2026-09-11 19:14:50,365 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Processing aliases funds,grouped,result
2026-09-11 19:14:50,365 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - detailed locations: M: funds[1,8],funds[-1,-1],result[15,9],grouped[13,10] C: result[15,9],grouped[13,10] R: result[15,9]
2026-09-11 19:14:50,374 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 0% complete
2026-09-11 19:14:50,374 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0008]
2026-09-11 19:15:25,406 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0008]
2026-09-11 19:15:25,486 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0008]
2026-09-11 19:15:48,339 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 11% complete
2026-09-11 19:15:48,342 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0008]
2026-09-11 19:15:57,367 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 15% complete
2026-09-11 19:15:57,367 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0008]
2026-09-11 19:16:16,520 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 21% complete
2026-09-11 19:16:16,523 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0008]
2026-09-11 19:16:24,249 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 26% complete
2026-09-11 19:16:24,256 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0008]
2026-09-11 19:16:39,628 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 33% complete
2026-09-11 19:16:39,629 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0008]
2026-09-11 19:16:51,315 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 37% complete
2026-09-11 19:16:51,315 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0008]
2026-09-11 19:16:54,375 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 42% complete
2026-09-11 19:16:54,375 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0008]
2026-09-11 19:17:09,275 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 46% complete
2026-09-11 19:17:09,275 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0008]
2026-09-11 19:17:22,636 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 52% complete
2026-09-11 19:17:22,640 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0008]
2026-09-11 19:17:36,759 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 58% complete
2026-09-11 19:17:36,760 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0008]
2026-09-11 19:17:41,899 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 63% complete
2026-09-11 19:17:41,902 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0008]

```

```

Administrator: Command Prompt
2026-09-11 19:17:49,151 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:17:49,224 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 100% complete
2026-09-11 19:17:49,232 [main] INFO org.apache.pig.tools.pigstats.mapreduce.SimplePigStats - Script Statistics:

HadoopVersion  PigVersion  UserId  StartedAt  FinishedAt  Features
3.2.4  0.18.0  ASUS  2026-09-11 19:14:49  2026-09-11 19:17:49  GROUP_BY

Success!

Job Stats (time in seconds):
JobId  Maps  Reduces  MaxMapTime  MinMapTime  AvgMapTime  MedianMapTime  MaxReduceTime  MinReduceTime  AvgReduceTime  MedianReduceTime  Alias  Feature Outputs
job_1789129568662_0008  34  5  31  8  19  18  104  7  65  65  funds,grouped,result  GROUP_BY,COMBINER  hdfs://localhost:9000/tmp/temp-1429656159/tmp-1804250032,

Input(s):
Successfully read 29033647 records (4459577292 bytes) from: "/mutual_funds/Mutual_Funds.csv"

Output(s):
Successfully stored 84 records (2980 bytes) in: "hdfs://localhost:9000/tmp/temp-1429656159/tmp-1804250032"

Counters:
Total records written : 84
Total bytes written : 2980
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1789129568662_0008

2026-09-11 19:17:49,237 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:17:49,241 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:17:49,321 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:17:49,326 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:17:49,369 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:17:49,383 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:17:49,413 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCARDED_TYPE_CONVERSION_FAILED 6686 time(s).
2026-09-11 19:17:49,413 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-11 19:17:49,419 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-11 19:17:49,420 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schema.tuple] was not set... will not generate code.
2026-09-11 19:17:49,442 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 5
2026-09-11 19:17:49,442 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 5
(1,11.987721049453354)
(Daily,361.8080345285863)
(Payout,11.33463102085789)
(466 Days,10.399010102301787)
(1100 days,11.820233600006645)
(1102 Days,12.845175444596473)
(5 Year Plan,11.2229304000072572)

```

```

Administrator: Command Prompt
(Equity Scheme - ELSS,76.87751301956895)
(Other Scheme - Gold ETF,2146.949884758513)
(Equity Scheme - Value Fund,52.47575132681864)
(Other Scheme - Other (FFs),479.5632400362651)
(Other Scheme - FoF Domestic,22.971199807631354)
(Equity Scheme - Large Cap Fund,55.47194338898238)
(Equity Scheme - Small Cap Fund,31.747749216634602)
(Hybrid Scheme - Arbitrage Fund,14.075733271637853)
(Debt Scheme - Corporate Bond Fund,178.5580131460975)
(Debt Scheme - Medium Duration Fund,21.28327115110448)
(Equity Scheme - Dividend Yield Fund,36.76241797199589)
(Hybrid Scheme - Multi Asset Allocation,30.642409484006887)
(Gilt,127.9559176549646)
(DIRECT,416.1536318164006)
(Direct,361.62685612015383)
(Income,65.1362484950151)
(1098 Days,11.698488115021268)
(1111 DAYS,12.399586666666666)
(1124 Days,10.617839181423492)
(1150 DAYS,10.796153683683681)
(Money Market,3.5417179944565778)
(erstwhile Cash Option,21.263720350597115)
(Debt Scheme - Dynamic Bond,61.53075257453476)
(Equity Scheme - Contra Fund,42.68189357890845)
(Equity Scheme - Focused Fund,11.748117682920845)
(Hybrid Scheme - Equity Savings,13.531922307271006)
(Debt Scheme - Money Market Fund,615.6047251903908)
(Hybrid Scheme - Balanced Hybrid Fund,45.86765829840074)
(Hybrid Scheme - Aggressive Hybrid Fund,51.44677164629662)
(Debt Scheme - Ultra Short Duration Fund,677.0478467928962)
(Hybrid Scheme - Conservative Hybrid Fund,36.600946103357799)
(Solution Oriented Scheme - Children's Fund,38.517093994404156)
(Hybrid Scheme - Dynamic Asset Allocation or Balanced Advantage,21.60030263446495)
(G,3.8256887955182033)
(IDF,154.0992981372648)
(ELSS,19.03116555676312)
(Balanced,26.282375968586381)
(1099 Days,11.487640751678407)
(1100 DAYS,13.768699999999999)
(1116 Days,10.766238148001564)
(1141 Days,11.383799088423066)
(1105 Days,10.190255080264203)
(54EB Growth,12.419509681554466)
(Annual Dividend,13.344722418879853)
(Scheme_Category,)
(Half Yearly Dividend,129.8793867899086)
(Compulsory Reinvestment,44.32023561103008)
(Debt Scheme - Floater Fund,150.72425703716934)
(Other Scheme - FoF Overseas,14.995621445052445)

```

```

Administrator: Command Prompt
(Debt Scheme - Gilt Fund,110.67173669745925)
(Debt Scheme - Liquid Fund,553.7419931106407)
(Other Scheme - Index Funds,32.23166386045864)
(Equity Scheme - Focused Fund,51.59925768227719)
(Formerly Super Institutional Plan,263.1954840570865)
(Formerly Known as IIFL Mutual Fund,375.7763347661777)
(Debt Scheme - Gilt Fund with 10 year constant duration,14.733313736145469)
(Liquid,1279.51516008342)
(91 Days,12.946533333333335)
(19 months Plan,19.24900274921882)
(Debt Scheme - Credit Risk Fund,91.56744432372462)
(Equity Scheme - Multi Cap Fund,59.195772051924614)
(Debt Scheme - Low Duration Fund,619.9133883278179)
(Debt Scheme - Long Duration Fund,43.64555792763532)
(Equity Scheme - Large & Mid Cap Fund,90.93638132377698)
(Debt Scheme - Medium to Long Duration Fund,21.400456420831837)
2026-09-11 19:17:49,749 [main] INFO org.apache.pig.Main - Pig script completed in 3 minutes, 1 second and 847 milliseconds (181847 ms)

C:\mutual_funds>pig>

```

Observed execution produced 84 category groups. Example outputs include Daily $\approx$ 361.8080, Payout $\approx$ 11.3346, 466 DAYS $\approx$ 10.3990, and 1100 days $\approx$ 11.8202.

7.4 Maximum NAV by Fund House

Objective: group records by Fund_House and identify the maximum NAV observed for each fund house.

```

funds = LOAD '/mutual_funds/Mutual_Funds.csv'
USING PigStorage(',')
AS (
Fund_House:chararray,
Scheme_Type:chararray,
Scheme_Category:chararray,

```

```

Scheme_Code:chararray,
Scheme_Name:chararray,
Date:chararray,
NAV:double
);

```

```
grouped_funds = GROUP funds BY Fund_House;
```

```
max_nav = FOREACH grouped_funds
```

```
GENERATE
```

```
group AS Fund_House,
```

```
MAX(funds.NAV) AS Maximum_NAV;
```

```
DUMP max_nav;
```

```

Administrator: Command Prompt
C:\mutual_funds\pig>notepad pig4_maxfundhouse.pig

C:\mutual_funds\pig>pig pig4_maxfundhouse.pig
2026-09-11 19:23:58,248 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-11 19:23:58,250 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-11 19:23:58,250 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-11 19:23:58,485 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-11 19:23:58,485 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig-1780133038402.log
2026-09-11 19:23:58,676 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\VASUS\pig\bootstrap not found
2026-09-11 19:23:58,710 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-11 19:23:58,710 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-11 19:23:58,998 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-pig4_maxfundhouse.pig-fb6c0255-ebb4-4c3d-9d8e-e0ad28a01baf
2026-09-11 19:23:58,998 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-11 19:23:59,471 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY
2026-09-11 19:23:59,485 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schemaTuple] was not set... will not generate code.
2026-09-11 19:23:59,509 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByCon
stParallelSetter, LimitOptimizer, LoadTypeCaster, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantC
alculator, SplitFilter, StreamTypeCaster]}
2026-09-11 19:23:59,543 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 6994800192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-11 19:23:59,573 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-11 19:23:59,583 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-11 19:23:59,600 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2026-09-11 19:23:59,600 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2026-09-11 19:23:59,646 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:23:59,821 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-11 19:23:59,831 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-11 19:23:59,834 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-11 19:23:59,834 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-11 19:23:59,841 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-11 19:23:59,842 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers
2026-09-11 19:23:59,843 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Using reducer estimator: org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputS
izeReducerEstimator
2026-09-11 19:23:59,848 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator - BytesPerReducer=1000000000 maxReducers=999 totalInputFileSize=4459429272
2026-09-11 19:23:59,848 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting Parallelism to 5
2026-09-11 19:23:59,848 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.reduce.tasks is deprecated. Instead, use mapreduce.job.reduces
2026-09-11 19:23:59,848 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - This job cannot be converted run in-process
2026-09-11 19:23:59,860 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.submit.replication is deprecated. Instead, use mapreduce.client.submit.file.replication
2026-09-11 19:24:00,013 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/pig-0.18.0-core-h3.jar to DistributedCache through /tmp/temp153155273
4/tmp2044822479/pig-0.18.0-core-h3.jar
2026-09-11 19:24:00,049 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/lib/automaton-1.11-8.jar to DistributedCache through /tmp/temp1531552
734/tmp-1873771915/automaton-1.11-8.jar
2026-09-11 19:24:00,083 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/lib/antlr-runtime-3.4.jar to DistributedCache through /tmp/temp153155
2734/tmp-268417946/antlr-runtime-3.4.jar
2026-09-11 19:24:00,118 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/lib/joda-time-2.14.0.jar to DistributedCache through /tmp/temp1531552
734/tmp-1771146721/joda-time-2.14.0.jar
2026-09-11 19:24:00,154 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/lib/RoaringBitmap-shaded-0.7.14.jar to DistributedCache through /tmp/
temp1531552734/tmp-192025746/RoaringBitmap-shaded-0.7.14.jar
2026-09-11 19:24:00,184 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting up single store job

```

```

Administrator: Command Prompt
2026-09-11 19:24:00,187 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Key [pig.schematuple] is false, will not generate code.
2026-09-11 19:24:00,188 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Starting process to move generated code to distributed cache
2026-09-11 19:24:00,188 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Setting key [pig.schematuple.classes] with classes to deserialize []
2026-09-11 19:24:00,226 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - 1 map-reduce Job(s) waiting for submission.
2026-09-11 19:24:00,231 [JobControl] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:24:00,239 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.task.id is deprecated. Instead, use mapreduce.task.attempt.id
2026-09-11 19:24:00,468 [JobControl] INFO org.apache.hadoop.mapreduce.JobResourceUploader - Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/ASUS/.staging/job_1789129568662_0009
2026-09-11 19:24:00,474 [JobControl] WARN org.apache.pig.builtin.PigStorage - Using PigTextInputFormat - No job jar file set. User classes may not be found. See Job or JobSetJar(String).
2026-09-11 19:24:00,501 [JobControl] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-11 19:24:00,501 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
2026-09-11 19:24:00,545 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths (combined) to process : 34
2026-09-11 19:24:00,606 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - number of splits:34
2026-09-11 19:24:00,629 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-11 19:24:00,696 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Submitting tokens for job: job_1789129568662_0009
2026-09-11 19:24:00,698 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Executing with tokens: []
2026-09-11 19:24:00,780 [JobControl] INFO org.apache.hadoop.mapred.YARNRunner - Job jar is not present. Not adding any jar to the list of resources.
2026-09-11 19:24:00,818 [JobControl] INFO org.apache.hadoop.conf.Configuration - resource-types.xml not found
2026-09-11 19:24:00,818 [JobControl] INFO org.apache.hadoop.yarn.util.resource.ResourceUtils - Unable to find 'resource-types.xml'.
2026-09-11 19:24:00,864 [JobControl] INFO org.apache.hadoop.yarn.client.api.impl.YarnClientImpl - Submitted application application_1789129568662_0009
2026-09-11 19:24:00,889 [JobControl] INFO org.apache.hadoop.mapreduce.Job - The url to track the job: http://NMU:8088/proxy/application_1789129568662_0009/
2026-09-11 19:24:00,889 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - HadoopJobId: job_1789129568662_0009
2026-09-11 19:24:00,889 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - Processing aliases funds,grouped,result
2026-09-11 19:24:00,889 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - detailed locations: M: funds[1,8],funds[-1,-1],result[15,9],grouped[13,10] C: result[15,9],gr
ouped[13,10] R: result[15,9]
2026-09-11 19:24:00,899 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - 0% complete
2026-09-11 19:24:00,899 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - Running jobs are [job_1789129568662_0009]
2026-09-11 19:24:38,263 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - 5% complete
2026-09-11 19:24:38,291 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - Running jobs are [job_1789129568662_0009]
2026-09-11 19:25:01,414 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - 10% complete
2026-09-11 19:25:01,421 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - Running jobs are [job_1789129568662_0009]
2026-09-11 19:25:08,580 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - 14% complete
2026-09-11 19:25:08,580 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - Running jobs are [job_1789129568662_0009]
2026-09-11 19:25:30,655 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - 21% complete
2026-09-11 19:25:30,657 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - Running jobs are [job_1789129568662_0009]
2026-09-11 19:25:35,951 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - 25% complete
2026-09-11 19:25:35,953 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - Running jobs are [job_1789129568662_0009]
2026-09-11 19:25:53,485 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - 31% complete
2026-09-11 19:25:53,497 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - Running jobs are [job_1789129568662_0009]
2026-09-11 19:26:10,971 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - 41% complete
2026-09-11 19:26:10,971 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - Running jobs are [job_1789129568662_0009]
2026-09-11 19:26:23,853 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - 47% complete
2026-09-11 19:26:23,853 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - Running jobs are [job_1789129568662_0009]
2026-09-11 19:26:38,690 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - 52% complete
2026-09-11 19:26:38,692 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - Running jobs are [job_1789129568662_0009]
2026-09-11 19:26:52,013 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - 57% complete
2026-09-11 19:26:52,013 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - Running jobs are [job_1789129568662_0009]
2026-09-11 19:26:58,708 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - 73% complete

```

```

Administrator: Command Prompt
2026-09-11 19:26:58,713 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - Running jobs are [job_1789129568662_0009]
2026-09-11 19:27:00,827 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - 89% complete
2026-09-11 19:27:00,860 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - Running jobs are [job_1789129568662_0009]
2026-09-11 19:27:01,909 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - Running jobs are [job_1789129568662_0009]
2026-09-11 19:27:07,038 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:27:07,055 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:27:07,421 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:27:07,439 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:27:07,502 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:27:07,517 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:27:07,616 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - 100% complete
2026-09-11 19:27:07,623 [main] INFO org.apache.pig.tools.pigstats.mapreduce.SimplePigStats - Script Statistics:

HadoopVersion  PigVersion  UserId  StartedAt  FinishedAt  Features
3.2.4  0.18.0  ASUS  2026-09-11 19:23:59  2026-09-11 19:27:07  GROUP_BY

Success!

Job Stats (time in seconds):
JobId  Maps  Reduces  MaxMapTime  MinMapTime  AvgMapTime  MedianMapTime  MaxReduceTime  MinReduceTime  AvgReduceTime  MedianReduceTime  Alias  Feature Outputs
job_1789129568662_0009  34  5  32  6  20  19  64  funds,grouped,result  GROUP_BY,COMBINER  hdfs://localhost:9000/tmp/temp1531552734/tmp1478457700,

Input(s):
Successfully read 29033647 records (4459577292 bytes) from: "/mutual_funds/Mutual_Funds.csv"

Output(s):
Successfully stored 71 records (2399 bytes) in: "hdfs://localhost:9000/tmp/temp1531552734/tmp1478457700"

Counters:
Total records written : 71
Total bytes written : 2399
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1789129568662_0009

2026-09-11 19:27:07,626 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:27:07,629 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:27:07,675 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:27:07,679 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:27:07,700 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:27:07,716 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:27:07,753 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - Encountered Warning FIELD_DISCARDED_TYPE_CONVERSION_FAILED 6606 time(s).
2026-09-11 19:27:07,753 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapreduce_layer.MapReduceLauncher - Success!
2026-09-11 19:27:07,760 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-11 19:27:07,762 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.

```

```

Administrator: Command Prompt
2026-09-11 19:27:07,753 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-11 19:27:07,760 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-11 19:27:07,762 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-11 19:27:07,780 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 5
2026-09-11 19:27:07,790 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 5
(DSP Mutual Fund,3285.6403)
(ING Mutual Fund,59.88)
(IDBI Mutual Fund,5635.3544)
(Tata Mutual Fund,4146.3089)
(Daiwa Mutual Fund,1382.3538)
(Groww Mutual Fund,2226.989)
(PPFAS Mutual Fund,1281.054)
(Quant Mutual Fund,99.1495)
(Fortis Mutual Fund,37.17)
(Bandhan Mutual Fund,2276.9581)
(Quantum Mutual Fund,2423.5152)
(Shinsei Mutual Fund,1079.6719)
(Benchmark Mutual Fund,2576.1705)
(PRINCIPAL Mutual Fund,2130.3541)
(Indiabulls Mutual Fund,2101.7877)
(PineBridge Mutual Fund,1811.8539)
(BNP Paribas Mutual Fund,3271.5995)
(Bajaj Finserv Mutual Fund,1003.0296)
(Canara Robeco Mutual Fund,3641.9971)
(Goldman Sachs Mutual Fund,3231.3129)
(Kotak Mahindra Mutual Fund,4645.3539)
(Franklin Templeton Mutual Fund,5225.7016)
(NI Mutual Fund,1060.7167)
(L&T Mutual Fund,3025.2376)
(SBI Mutual Fund,5276.5905)
(Axis Mutual Fund,4456.3444)
(HSBC Mutual Fund,3208.0012)
(Navi Mutual Fund,2411.5787)
(Essex Mutual Fund,1447.8284)
(Baroda Mutual Fund,2219.1396)
(Taurus Mutual Fund,2868.2686)
(Invesco Mutual Fund,5468.2761)
(JPMorgan Mutual Fund,35.936)
(Sundaram Mutual Fund,6495.5652)
(Bank of India Mutual Fund,2861.7319)
(Baroda Pioneer Mutual Fund,1668.405)
(ITI Mutual Fund,1282.2216)
(IDFC Mutual Fund,1777010.5088)
(Fidelity Mutual Fund,39.315)
(DHFL Pramerica Mutual Fund,1631.8015)
(Aditya Birla Sun Life Mutual Fund,5114.958)
(Fund House)
(LIC Mutual Fund,4174.6559)
(UTI Mutual Fund,102410.0)

```

```

Administrator: Command Prompt
(IIFL Mutual Fund,1056.0332)
(Samco Mutual Fund,1050.513)
(Union Mutual Fund,2216.7056)
(Sahara Mutual Fund,3348.7724)
(Shriram Mutual Fund,27.6952)
(Deutsche Mutual Fund,208.4716)
(Edelweiss Mutual Fund,19631.9267)
(Nippon India Mutual Fund,5624.7124)
(Standard Chartered Mutual Fund,1173.3449)
(YES Mutual Fund,1003.4952)
(HDFC Mutual Fund,5548.09)
(Trust Mutual Fund,1118.9727)
(quant Mutual Fund,515.8658)
(BOI AXA Mutual Fund,1597.022)
(Reliance Mutual Fund,3166.5923)
(ABN AMRO Mutual Fund,48.12)
(DBS Chola Mutual Fund,53.21)
(Principal Mutual Fund,3231.0113)
(PGIM India Mutual Fund,2520.1955)
(Mirae Asset Mutual Fund,2428.4098)
(JM Financial Mutual Fund,1161.7975)
(Motilal Oswal Mutual Fund,3373.2839)
(Morgan Stanley Mutual Fund,1288.4008)
(ICICI Prudential Mutual Fund,1918470.3931)
(WhiteOak Capital Mutual Fund,1234.4983)
(Mahindra Manulife Mutual Fund,1510.2299)
(Baroda BNP Paribas Mutual Fund,3864.422)
2026-09-11 19:27:08,099 [main] INFO org.apache.pig.Main - Pig script completed in 3 minutes, 9 seconds and 861 milliseconds (189861 ms)

C:\mutual_funds>pig>_

```

Observed execution produced 71 Fund House groups. Example maximum-NAV values include DSP Mutual Fund = 3285.6403, IDBI Mutual Fund = 5635.3544, Tata Mutual Fund = 4146.3089, Franklin Templeton Mutual Fund = 5225.7016, ICICI Prudential Mutual Fund = 1918470.3931, and UTI Mutual Fund = 102410.0.

7.5 Scheme-Type Record Count

Objective: group records by Scheme_Type and count the number of records in every group.

```
funds = LOAD '/mutual_funds/Mutual_Funds.csv'
```

```
USING PigStorage(',')
```

```
AS (
```

```
Fund_House:chararray,
```

```
Scheme_Type:chararray,
```

```
Scheme_Category:chararray,
```

```
Scheme_Code:chararray,
```

```
Scheme_Name:chararray,
```

```
Date:chararray,
```

```
NAV:double
```

```
);
```

```
grouped_type = GROUP funds BY Scheme_Type;
```

```
scheme_count = FOREACH grouped_type
```

```
GENERATE
```

```
group AS Scheme_Type,
```

```
COUNT(funds) AS Total_Records;
```

```
DUMP scheme_count;
```

```
Administrator: Command Prompt

C:\mutual_funds\pig>notepad pig5_schemetype.pig

C:\mutual_funds\pig>pig pig5_schemetype.pig
2026-09-11 19:31:13,938 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-11 19:31:13,940 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-09-11 19:31:13,940 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecType
2026-09-11 19:31:14,092 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-11 19:31:14,093 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig-1789133474089.log
2026-09-11 19:31:14,383 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\ASUS/.pigbootstrap not found
2026-09-11 19:31:14,419 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.job.tracker.address
2026-09-11 19:31:14,419 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: hdfs://localhost:9000
2026-09-11 19:31:14,788 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-pig5_schemetype.pig-4836af03-fa2a-4abd-97f8-06f68c6c17171
2026-09-11 19:31:14,788 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-11 19:31:15,216 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY
2026-09-11 19:31:15,230 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2026-09-11 19:31:15,258 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - [RULES_ENABLED]=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByCon
tainerOptimizer, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantC
alculator, SplitFilter, StreamTypeCastInserter]]
2026-09-11 19:31:15,294 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-11 19:31:15,329 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-11 19:31:15,344 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-11 19:31:15,361 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2026-09-11 19:31:15,361 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2026-09-11 19:31:15,406 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:31:15,599 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-11 19:31:15,608 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScripState - Pig script settings are added to the job
2026-09-11 19:31:15,612 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-11 19:31:15,612 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-11 19:31:15,615 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-11 19:31:15,618 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers.
2026-09-11 19:31:15,619 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Using reducer estimator: org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputS
izeReducerEstimator
2026-09-11 19:31:15,624 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator - BytesPerReducer=1000000000 maxReducers=999 totalInputFileSize=4459429272
2026-09-11 19:31:15,624 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting Parallelism to 5
2026-09-11 19:31:15,624 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.reduce.tasks is deprecated. Instead, use mapreduce.job.reduces
2026-09-11 19:31:15,624 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - This job cannot be converted run in-process
2026-09-11 19:31:15,635 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.submit.replication is deprecated. Instead, use mapreduce.client.submit.file.replication
2026-09-11 19:31:15,764 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/pig-0.18.0-core-h3.jar to DistributedCache through /tmp/temp210405314
0/tmp-78319860/pig-0.18.0-core-h3.jar
2026-09-11 19:31:15,788 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/lib/automaton-1.11-8.jar to DistributedCache through /tmp/temp2104053
140/tmp-1901521748/automaton-1.11-8.jar
2026-09-11 19:31:15,813 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/lib/antlr-runtime-3.4.jar to DistributedCache through /tmp/temp210405
3140/tmp-1983707137/antlr-runtime-3.4.jar
2026-09-11 19:31:15,828 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/lib/joda-time-2.14.0.jar to DistributedCache through /tmp/temp2104053
140/tmp-378346339/joda-time-2.14.0.jar
2026-09-11 19:31:15,860 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/C:/pig/lib/RoaringBitmap-shaded-0.7.14.jar to DistributedCache through /tmp/
temp2104053140/tmp-219646911/RoaringBitmap-shaded-0.7.14.jar
2026-09-11 19:31:15,898 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting up single store job
```

```

Administrator: Command Prompt
2026-09-11 19:31:15,903 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Key [pig.schematuple] is false, will not generate code.
2026-09-11 19:31:15,903 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Starting process to move generated code to distributed cache
2026-09-11 19:31:15,903 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Setting key [pig.schematuple.classes] with classes to deserialize []
2026-09-11 19:31:15,951 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 1 map-reduce Job(s) waiting for submission.
2026-09-11 19:31:15,958 [JobControl] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:31:15,974 [JobControl] INFO org.apache.hadoop.mapred.conf.Configuration.deprecation - mapred.task.id is deprecated. Instead, use mapreduce.task.attempt.id
2026-09-11 19:31:16,323 [JobControl] INFO org.apache.hadoop.mapreduce.JobResourceUploader - Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/ASUS/.staging/job_1789129568662_0010
2026-09-11 19:31:16,330 [JobControl] WARN org.apache.pig.builtin.PigStorage - Using PigTextInputFormat - No job jar file set. User classes may not be found. See Job or JobsetJar(String).
2026-09-11 19:31:16,385 [JobControl] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2026-09-11 19:31:16,385 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
2026-09-11 19:31:16,406 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths (combined) to process : 34
2026-09-11 19:31:16,470 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - number of splits:34
2026-09-11 19:31:16,525 [JobControl] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2026-09-11 19:31:16,582 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Submitting tokens for job: job_1789129568662_0010
2026-09-11 19:31:16,584 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Executing with tokens: []
2026-09-11 19:31:16,658 [JobControl] INFO org.apache.hadoop.mapred.YARNRunner - Job jar is not present. Not adding any jar to the list of resources.
2026-09-11 19:31:16,696 [JobControl] INFO org.apache.hadoop.conf.Configuration - resource-types.xml not found
2026-09-11 19:31:16,708 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Unable to find 'resource-types.xml'.
2026-09-11 19:31:16,742 [JobControl] INFO org.apache.hadoop.yarn.client.api.impl.YarnClientImpl - Submitted application application_1789129568662_0010
2026-09-11 19:31:16,767 [JobControl] INFO org.apache.hadoop.mapreduce.Job - The url to track the job: http://MO:8088/proxy/application_1789129568662_0010/
2026-09-11 19:31:16,767 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - HadoopJobId: job_1789129568662_0010
2026-09-11 19:31:16,767 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Processing aliases funds,grouped,result
2026-09-11 19:31:16,767 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - detailed locations: M: funds[1,8],funds[-1,-1],result[15,9],grouped[13,10] C: result[15,9],gr
ouped[13,10] R: result[15,9]
2026-09-11 19:31:16,778 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 0% complete
2026-09-11 19:31:16,778 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0010]
2026-09-11 19:31:51,239 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 5% complete
2026-09-11 19:31:51,246 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0010]
2026-09-11 19:32:11,986 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 11% complete
2026-09-11 19:32:13,989 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0010]
2026-09-11 19:32:21,342 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 15% complete
2026-09-11 19:32:21,348 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0010]
2026-09-11 19:32:39,267 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 19% complete
2026-09-11 19:32:46,423 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0010]
2026-09-11 19:32:46,423 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 24% complete
2026-09-11 19:32:46,445 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0010]
2026-09-11 19:33:04,410 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 32% complete
2026-09-11 19:33:04,415 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0010]
2026-09-11 19:33:24,347 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 40% complete
2026-09-11 19:33:41,378 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0010]
2026-09-11 19:33:41,386 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 46% complete
2026-09-11 19:34:01,450 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0010]
2026-09-11 19:34:01,453 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 52% complete
2026-09-11 19:34:21,592 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0010]
2026-09-11 19:34:29,322 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 63% complete

```

```

Administrator: Command Prompt
2026-09-11 19:34:37,348 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0010]
2026-09-11 19:34:37,348 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 89% complete
2026-09-11 19:34:37,360 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0010]
2026-09-11 19:34:37,360 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0010]
2026-09-11 19:34:37,393 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0010]
2026-09-11 19:34:37,393 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789129568662_0010]
2026-09-11 19:34:37,777 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:34:37,783 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:34:37,783 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:34:37,844 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:34:37,851 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:34:37,943 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 100% complete
2026-09-11 19:34:37,952 [main] INFO org.apache.pig.tools.pigstats.mapreduce.SimplePigStats - Script Statistics:

HadoopVersion PigVersion UserId StartedAt FinishedAt Features
3.2.4 0.18.0 ASUS 2026-09-11 19:31:15 2026-09-11 19:34:37 GROUP_BY

Success!

Job Stats (time in seconds):
JobId Maps Reduces MaxMapTime MinMapTime AvgMapTime MedianMapTime MaxReduceTime MinReduceTime AvgReduceTime MedianReduceTime Alias Feature Outputs
job_1789129568662_0010 34 5 30 10 21 19 132 10 70 86 funds,grouped,result hdfs://localhost:9000/tmp/2104053140/tmp742618004,

Input(s):
Successfully read 29033647 records (445957292 bytes) from: "/mutual_funds/Mutual_Funds.csv"

Output(s):
Successfully stored 78 records (5326 bytes) in: "hdfs://localhost:9000/tmp/2104053140/tmp742618004"

Counters:
Total records written : 78
Total bytes written : 5326
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1789129568662_0010

2026-09-11 19:34:37,957 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:34:37,967 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:34:38,043 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:34:38,047 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:34:38,087 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-11 19:34:38,097 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-11 19:34:38,136 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCARDED_TYPE_CONVERSION_FAILED 6606 time(s).
2026-09-11 19:34:38,144 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-11 19:34:38,144 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.

```

```
Administrator: Command Prompt
2026-09-11 19:34:38,186 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 5
(IIFLS Mutual Fund,2985265)
(128471;INF192K01EAF;-;JM Short Term Fund,7888)
(14248;INF200K01K5;-;SBI DEBT FUND SERIES C - 9,888)
(135174;INF192K01K00;INF192K01K88;JM Arbitrage Fund,1712)
(119362;INF955L01BQ7;-;BARODA PIONEER INCOME FUND - Plan B,7674)
(139259;INF192K01C8;-;JM Liquid Fund Unclaimed Redemption IEF,29221)
(134130;INF955L01G13;-;Baroda Pioneer Hybrid Fund- Series 1- Plan B,10134)
(134009;INF955L01D14;-;Baroda Pioneer Treasury Advantage Fund - Plan B,998)
(119390;INF955L01A04;-;BARODA PIONEER CONSERVATIVE HYBRID FUND - Plan B,186)
(129210;INF903J01F76;INF903J01F84;Sundaram Infrastructure Advantage Fund,8450)
(144748;INF209KB1V58;-;Aditya Birla Sun Life Fixed Term Plan - Series QY,30443)
(138559;INF663L01K04;-;DHFL Pramerica Hybrid Fixed Term Fund - Series 11 - Direct Plan - Dividend,38)
(Close Ended Schemes,629289)
(142400;INF740K01E5;-;DGP A.C.E. Fund,7318)
(134818;INF174K01R71;-;Kotak FMP Series 175,7252)
(143611;-;INF192K01D07;JM Low Duration Fund,10159)
(120440;INF192K01D1;-;INF192K01D8;JM MIP Fund,11188)
(101281;INF179K01L09;-;HDFC Index FundSensex Plan,9917)
(123970;INF846K01Z09;-;Axis Hybrid Fund - Series 7,1468)
(131466;INF200K01100;-;SBI DEBT FUND SERIES A - 44,15)
(130983;INF200K010P2;-;SBI DEBT FUND SERIES A - 39,2)
(134045;INF955L01G2;-;Baroda Pioneer ELSS 96 Plan B,472)
(120455;-;INF192K01D04;JM Ultra Short Duration Fund -,307)
(14152;INF194K01A02;-;IDFC EQUITY OPPORTUNITY SERIES 6,25340)
(144833;INF846K01H48;-;Axis Fixed Term Plan - Series 06,6744)
(126788;INF789F81T83;INF789F81T1C1;UTI FTIF Series XVII -IX,782)
(133258;INF789F17T4;-;UTI Focussed Equity Fund - Series II,2924)
(119326;INF955L01948;-;BARODA PIONEER HYBRID EQUITY FUND - Plan B,4053)
(103394;-;INF903J01G06;Sundaram Money Fund Regular weekly Dividend. Rein,10516)
(138534;INF223J01669;-;DHFL Pramerica Hybrid Fixed Term Fund - Series 4 - Dividend,8867)
(138940;INF223J01A289;-;DHFL Pramerica Mid Cap Fund - Series 1 - Direct Plan - Dividend,440)
(113483;-;INF251K01D08;BNP PARIBAS LOW DURATION Fund-Regular Plan- Daily Dividend Option,1522)
(138988;INF223J01A04;-;DHFL Pramerica Large Cap Fund - Series 1 - Regular Plan - Dividend,629)
(Scheme_Type,1)
(Interval Fund Schemes,150167)
(120492;INF192K01C7;-;JM Multicap Fund,4391)
(133662;INF174K01G82;-;Kotak FMP Series 171,2968)
(135488;INF174K01Y15;-;Kotak FMP Series 180,396365)
(103182;-;Aditya Birla Sun Life Income Fund-Plan G,14822)
(125254;INF174K01V07;INF174K01V55;Kotak Bond Short Term Plan-,47684)
(119317;INF955L01A05;-;Baroda Pioneer Dynamic Bond Fund - Plan B,13473)
(112659;INF090T01403;-;Franklin India Fixed Tenure Fund - Series XII - Plan C,44092)
(119333;INF955L01971;-;Baroda Pioneer Banking and Financial Services Fund - Plan B,1642)
(105556;INF903J01J09;INF903J01E05;Sundaram Low Duration Fund Regular Weekly Dividend,2567)
(139020;INF223J01F75;-;DHFL Pramerica Large Cap Fund - Series 3 - Direct Plan - Dividend,652)
(138586;INF223J01T05;-;DHFL Pramerica Hybrid Fixed Term Fund - Series 13 - Regular Plan - Dividend,823)
(138655;INF663L01S06;-;DHFL Pramerica Hybrid Fixed Term Fund - Series 19 - Regular Plan - Dividend,26256)
(IIFCL Mutual Fund,64)

(138586;INF223J01TH5;-;DHFL Pramerica Hybrid Fixed Term Fund - Series 13 - Regular Plan - Dividend,823)
(138655;INF663L01S06;-;DHFL Pramerica Hybrid Fixed Term Fund - Series 19 - Regular Plan - Dividend,26256)
(IIFCL Mutual Fund,64)
(Open Ended Schemes,15162950)
(360 ONE Mutual Fund,3532025)
(135269;INF789F81V43;-;UTI FTIF Series XXII-XIV,2696)
(134664;INF192K01JY6;INF192K01KB2;JM G-Sec Fund -,3071)
(130158;INF200K01G12;-;SBI Debt Fund Series A - 29,3)
(141581;INF179KB1V39;-;HDFC EOF - II - 1100D June 2017,12292)
(144875;INF846K01I21;-;Axis Fixed Term Plan - Series 97,10108)
(133524;INF955L01GA0;-;Baroda Pioneer Credit Risk Fund -Plan B-,496)
(138536;INF223J01A00;-;DHFL Pramerica Hybrid Fixed Term Fund - Series 5 - Dividend,503)
(139018;INF223J01A1F57;-;DHFL Pramerica Large Cap Fund - Series 3 - Regular Plan - Dividend,652)
(138555;INF223J01MG2;-;DHFL Pramerica Hybrid Fixed Term Fund - Series 10 - Annual Dividend,380)
(138581;INF663L01LL7;-;DHFL Pramerica Hybrid Fixed Term Fund - Series 12 - Regular Plan - Dividend,503)
(138600;INF223J01U25;-;DHFL Pramerica Hybrid Fixed Term Fund - Series 14 - Regular Plan - Dividend,580)
(101997;-;HDFC Overnight Fund - Dividend option,3988)
(133391;INF192K01IX0;-;JM Floater Short Term Fund -,9422)
(119341;INF955L01A07;-;BARODA PIONEER GILT FUND - Plan B,14019)
(141923;INF179KB1W87;-;HDFC HOF - I - 1140D November 2017,19172)
(110594;#N/A;-;CANARA ROBEKO FIXED MATURITY PLAN- Series 4,39684)
(119416;INF955L01A06;-;BARODA PIONEER LIQUID FUND - PLAN B,56540)
(100042;INF209K01ME4;-;Aditya Birla Sun Life Liquid Fund-Retail,186)
(102055;INF200K01743;-;SBI Magnum Income Plus Fund - Savings Plan,3570)
(100062;-;INF209K01D07;Aditya Birla Sun Life Gilt Plus-Regular Plan,339)
(140463;INF955L01G08;-;Baroda Pioneer Short Term Bond Fund - Plan B -,10010)
(101221;INF109K01TW3;-;ICICI Prudential Money Market Fund Retail Growth,6275)
(132800;INF205K01L62;-;Invesco India Fixed Maturity Plan-Series 24-Plan F,18101)
(134059;INF205K01S65;-;Invesco India Fixed Maturity Plan - Series 25 - Plan D,8718)
(138537;INF223J01L85;-;DHFL Pramerica Hybrid Fixed Term Fund - Series 6 - Dividend,644)
(138951;INF223J01A1263;-;DHFL Pramerica Mid Cap Fund - Series 1 - Regular Plan - Dividend,440)
(138985;INF223J01A1A0;-;DHFL Pramerica Large Cap Fund - Series 1 - Direct Plan - Dividend,629)
(138652;INF663L01S06;-;DHFL Pramerica Hybrid Fixed Term Fund - Series 17 - Regular Plan - Dividend,523)
2026-09-11 19:34:38,536 [main] INFO org.apache.pig.Main - Pig script completed in 3 minutes, 24 seconds and 587 milliseconds (204587 ms)

C:\mutual_funds>pig>
```

Observed execution produced 78 output groups. The result included major values such as Open Ended Schemes = 15,162,950, Close Ended Schemes = 626,289, Interval Fund Schemes = 150,167, and additional scheme-type-like values present in the raw dataset.

8. Project Repository

GitHub repository: <https://github.com/Mou-Chakraborty/Matual-funds-Data-Analysis-by-Hadoop>

9. Reference

Siddhi Prabhu, “YouTube Data Analysis using Hadoop,” GitHub repository:
<https://github.com/SiddhiPrabhu1995/YouTube-Data-Analysis-using-Hadoop>